

BFS Traversal

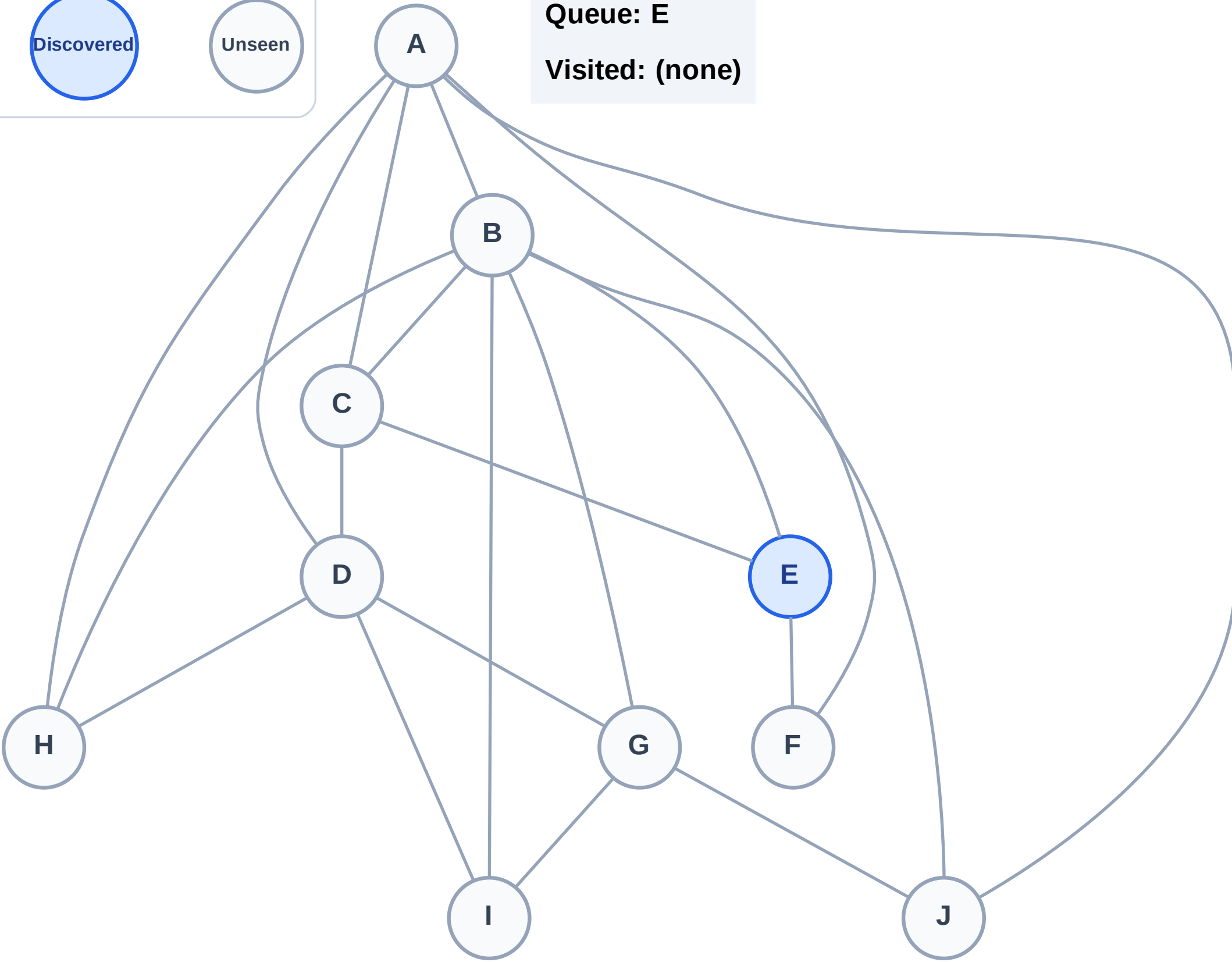
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

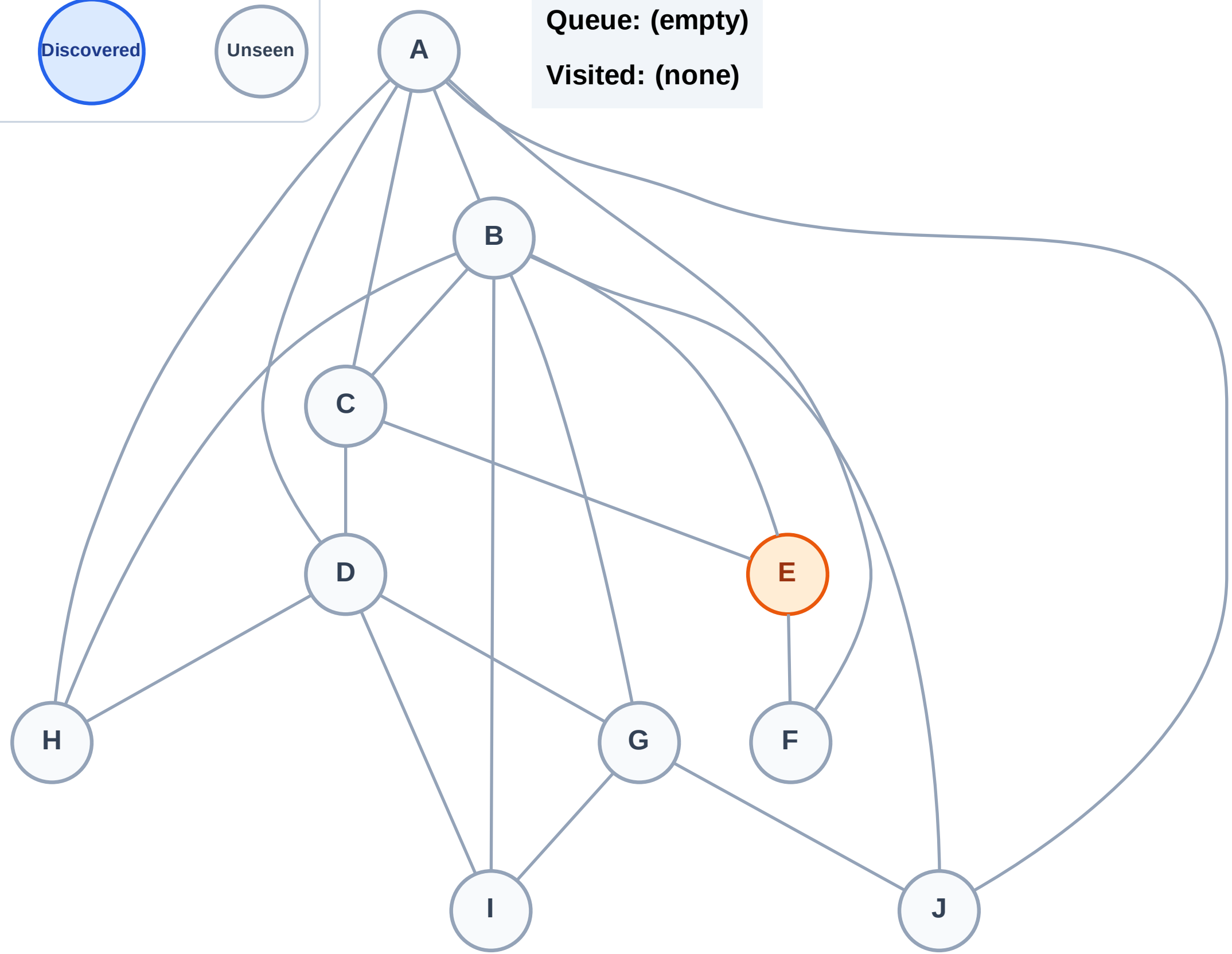
Step 2: Process node E

Legend



Queue: (empty)

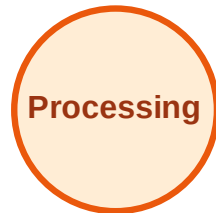
Visited: (none)



BFS Traversal

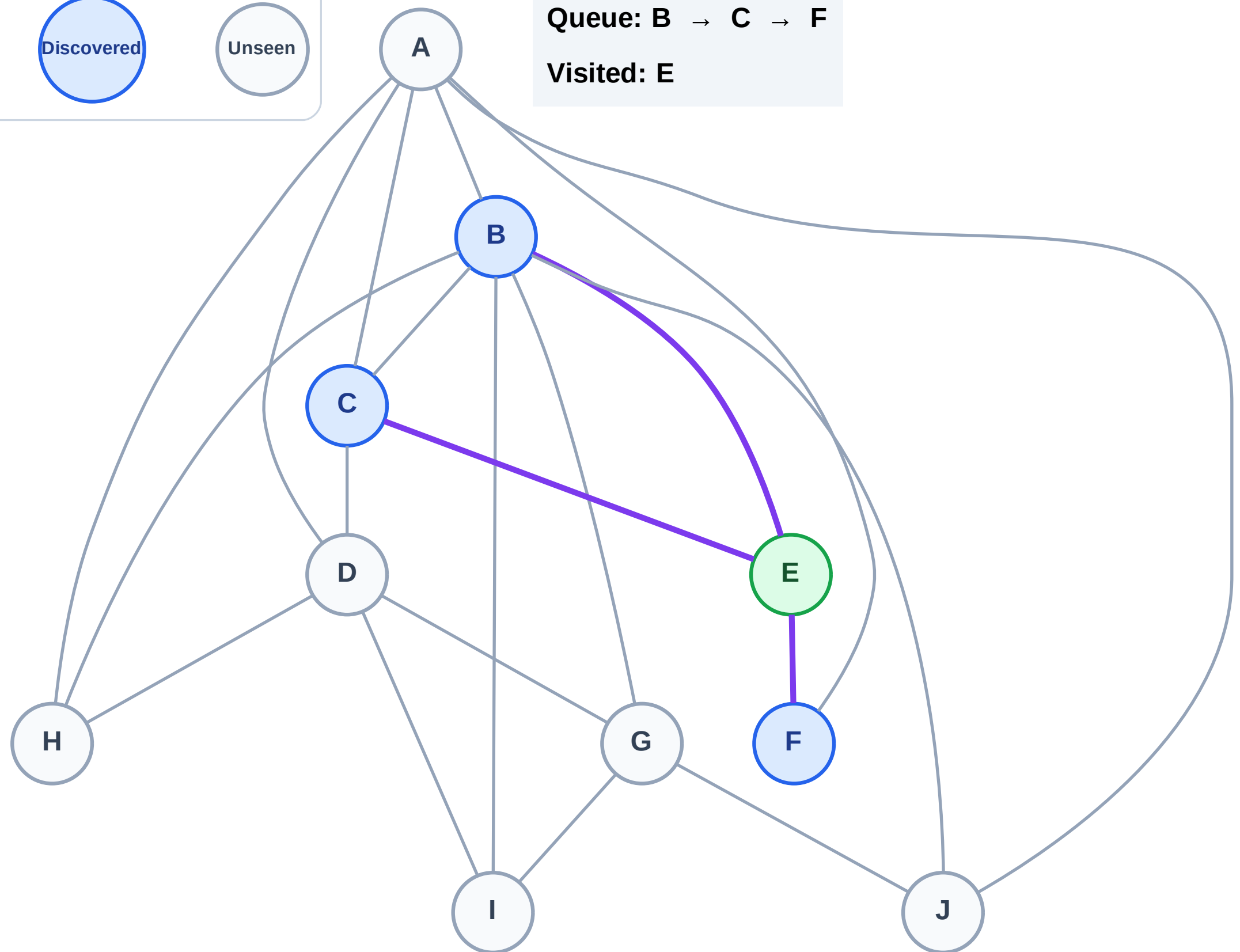
Step 3: Discover B, C, F from E

Legend



Queue: B → C → F

Visited: E



BFS Traversal

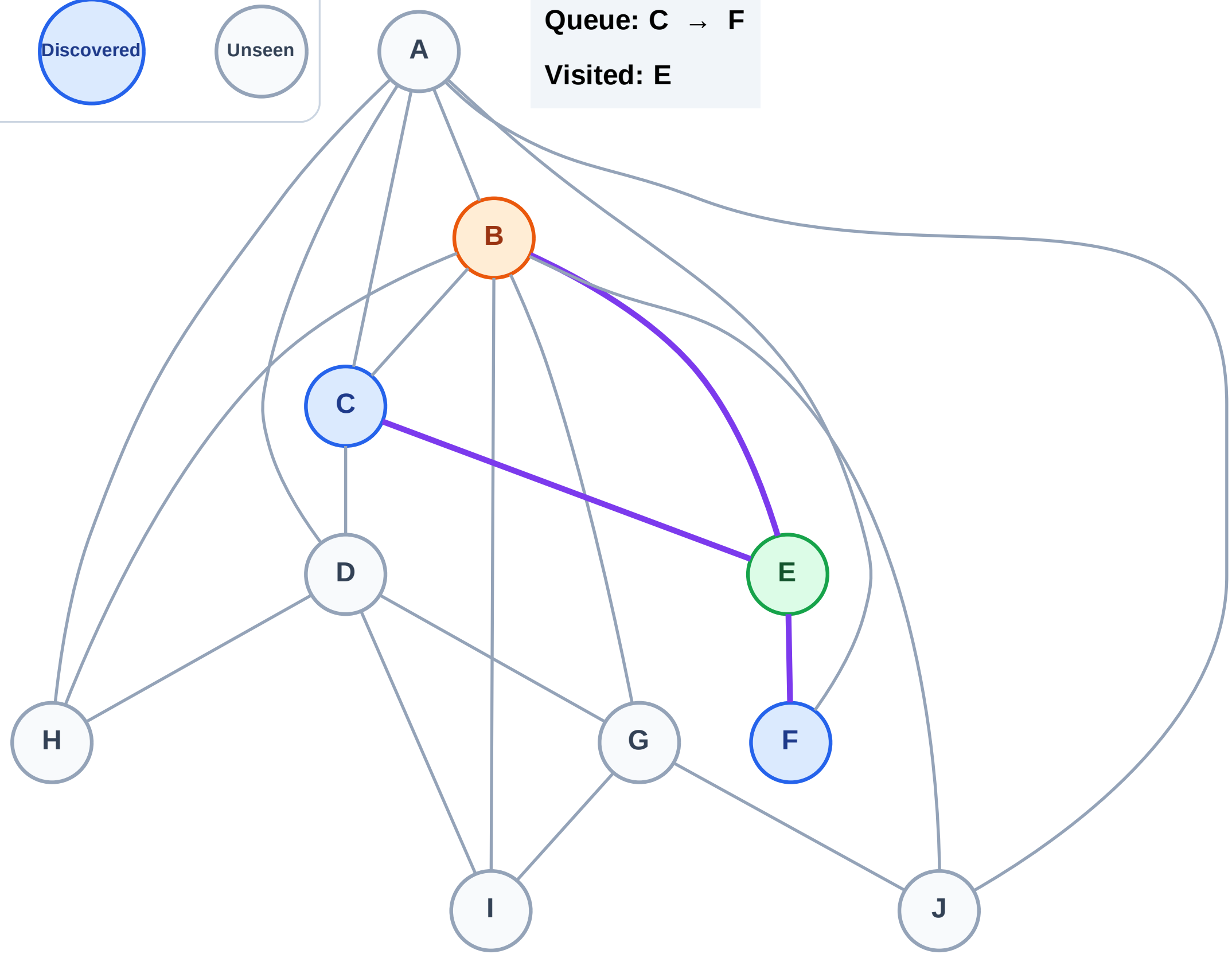
Step 4: Process node B

Legend



Queue: C → F

Visited: E



BFS Traversal

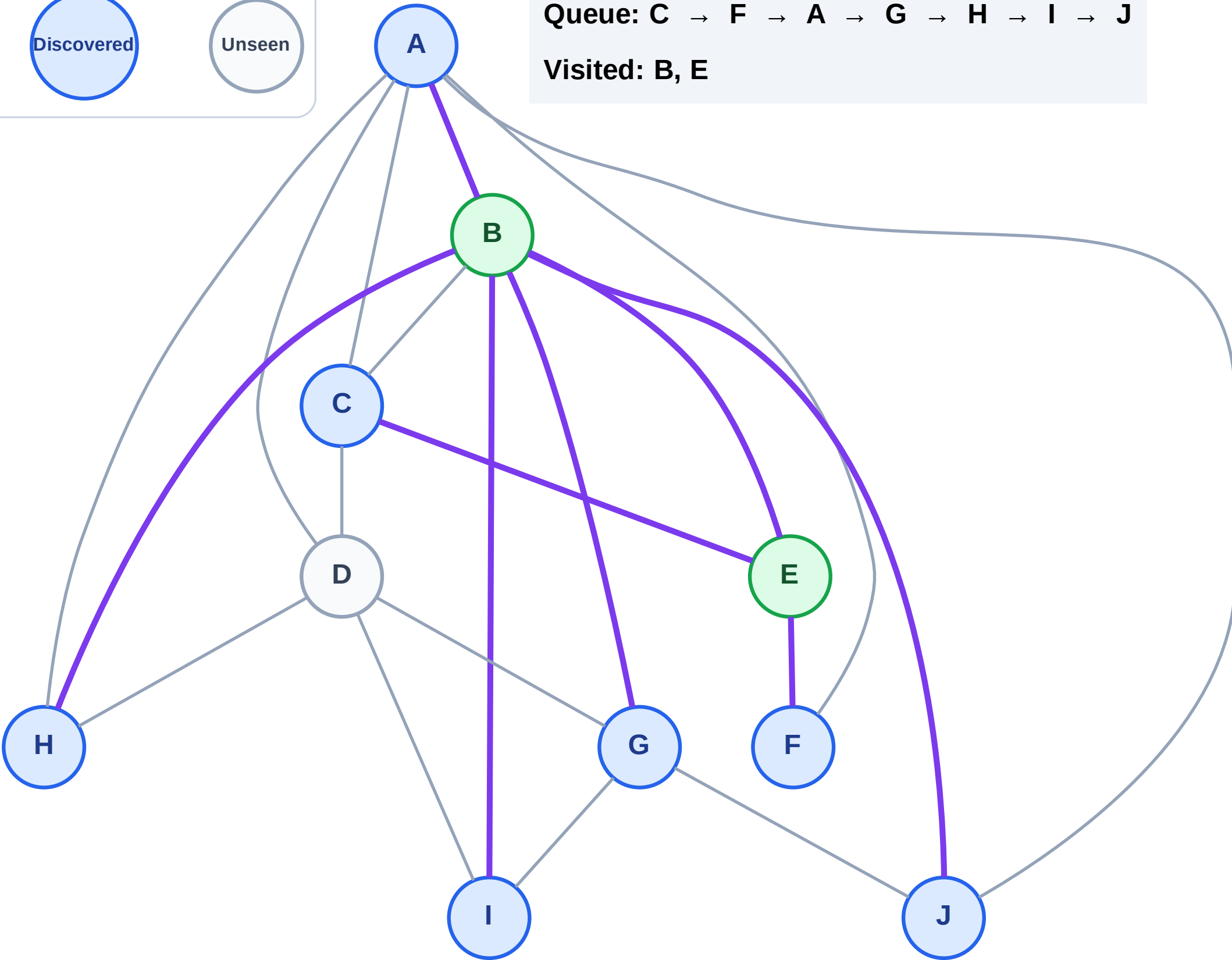
Step 5: Discover A, G, H, I, J from B

Legend



Queue: C → F → A → G → H → I → J

Visited: B, E



BFS Traversal

Step 6: Process node C

Legend

Complete

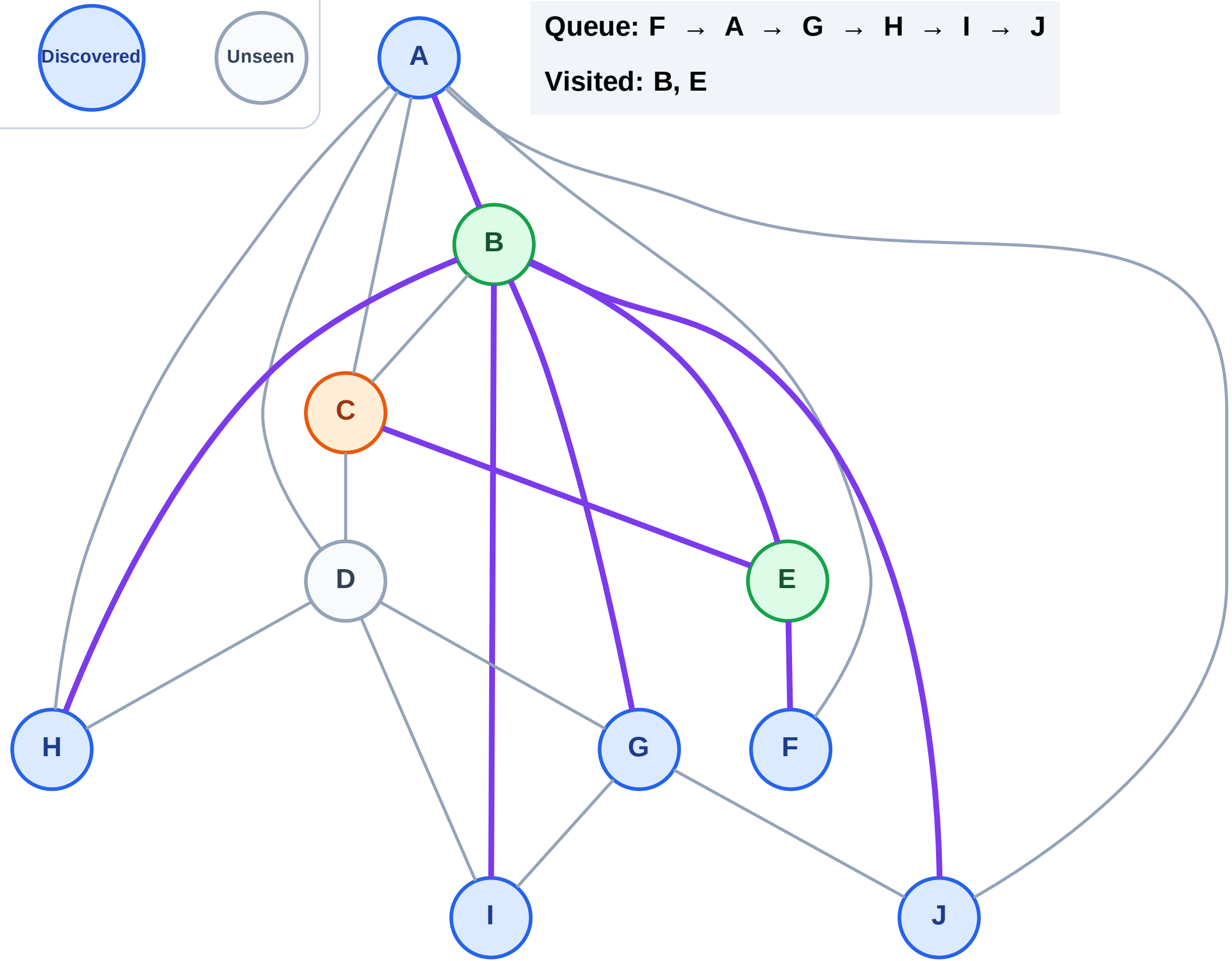
Processing

Discovered

Unseen

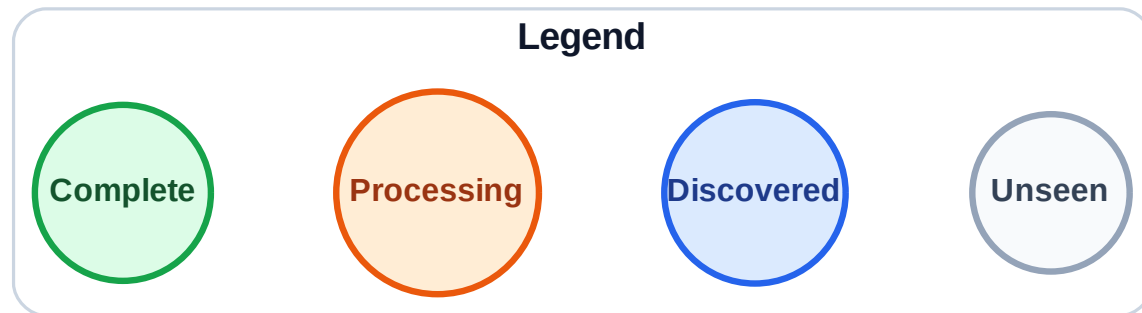
Queue: F → A → G → H → I → J

Visited: B, E



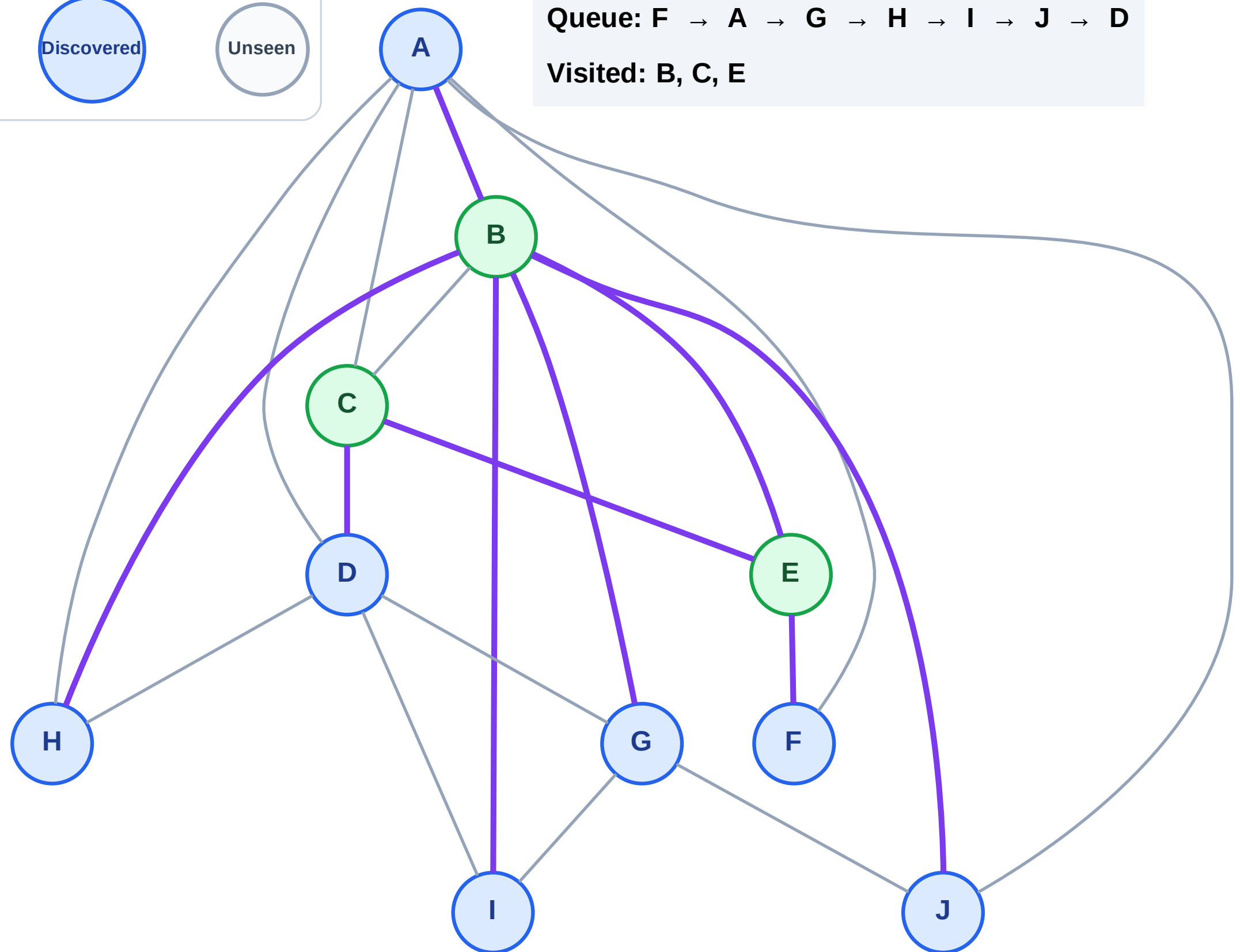
BFS Traversal

Step 7: Discover D from C



Queue: F → A → G → H → I → J → D

Visited: B, C, E



BFS Traversal

Step 8: Process node F

Legend

Complete

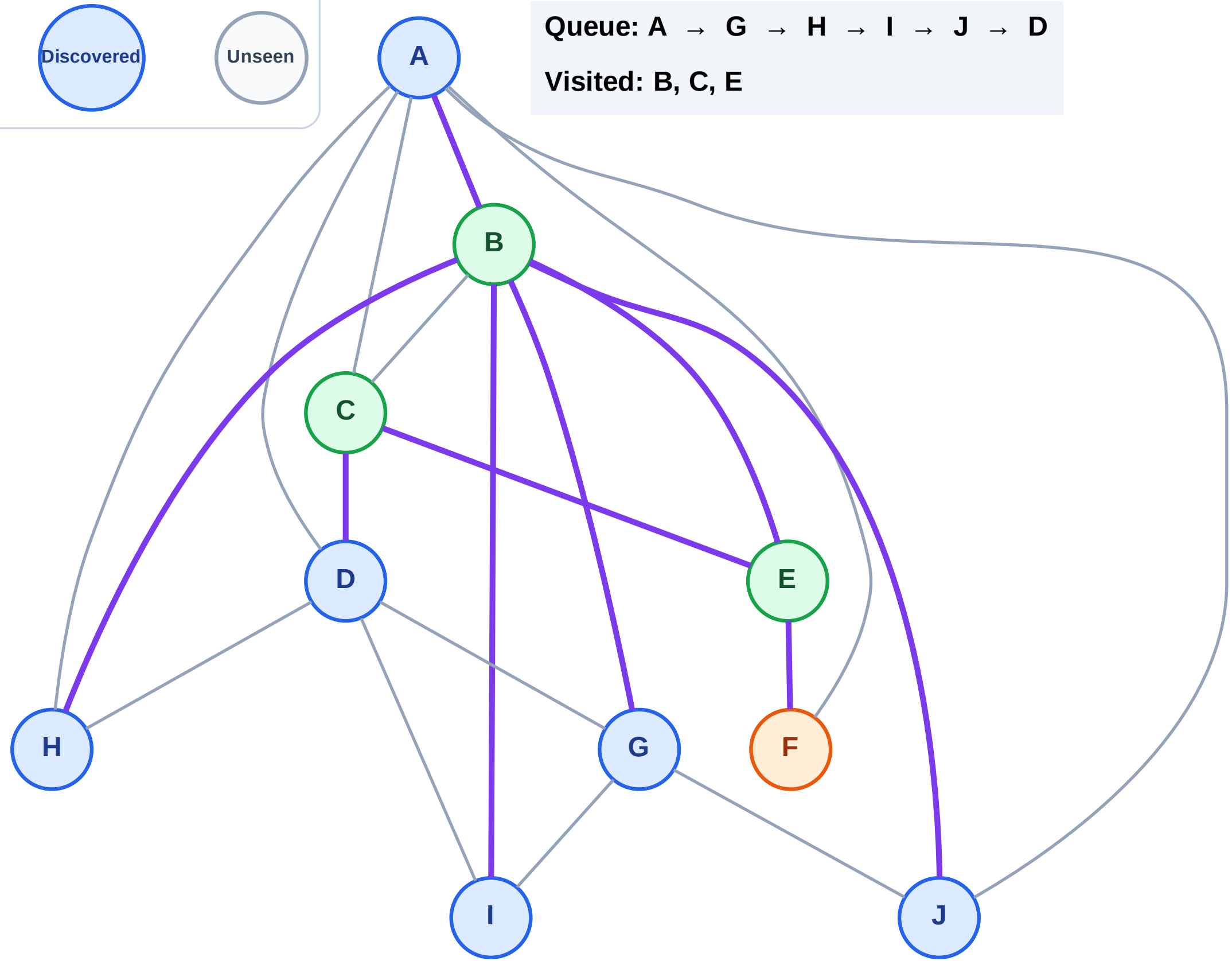
Processing

Discovered

Unseen

Queue: A → G → H → I → J → D

Visited: B, C, E



BFS Traversal

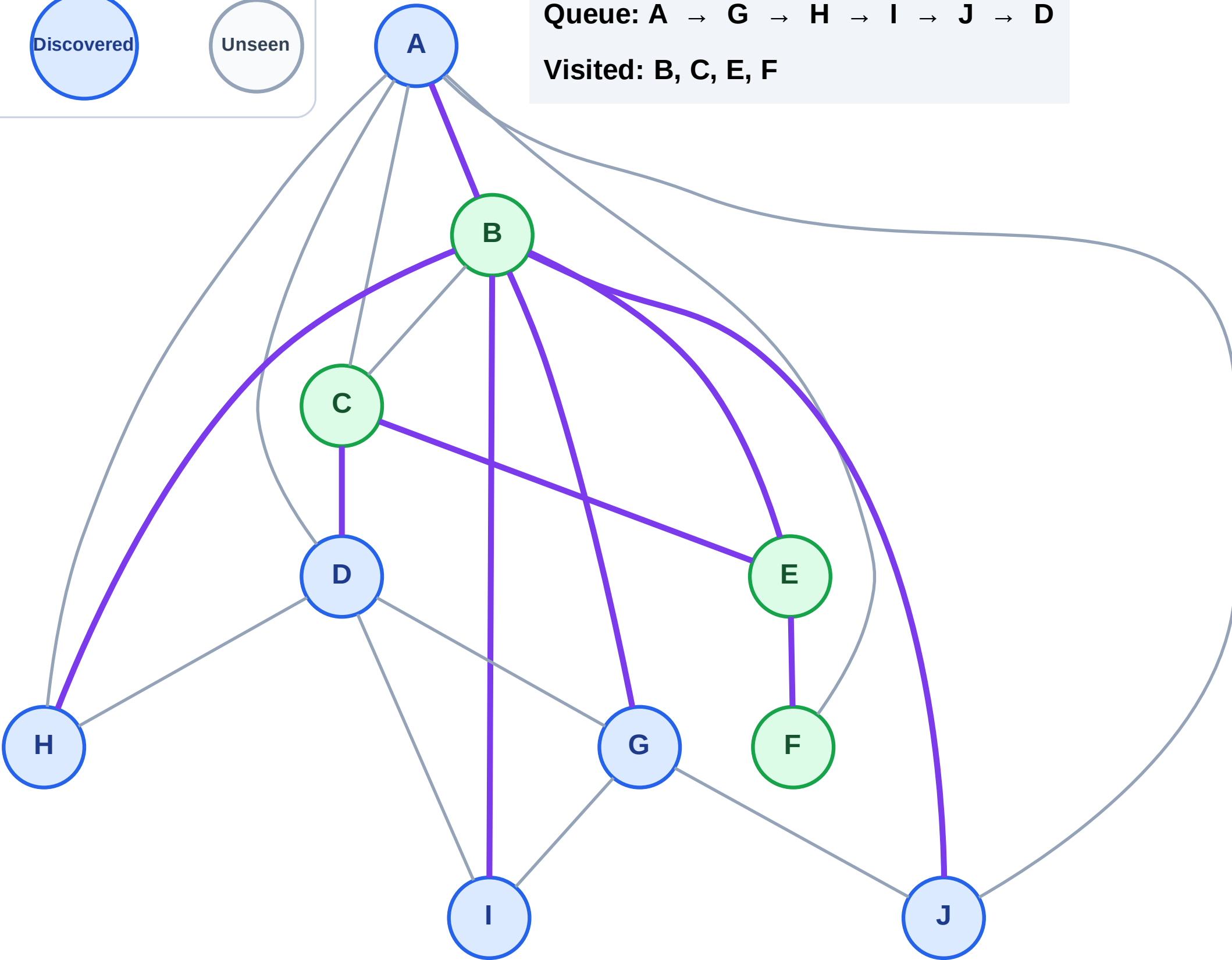
Step 9: No new neighbors from F

Legend



Queue: A → G → H → I → J → D

Visited: B, C, E, F



BFS Traversal

Step 10: Process node A

Legend

Complete

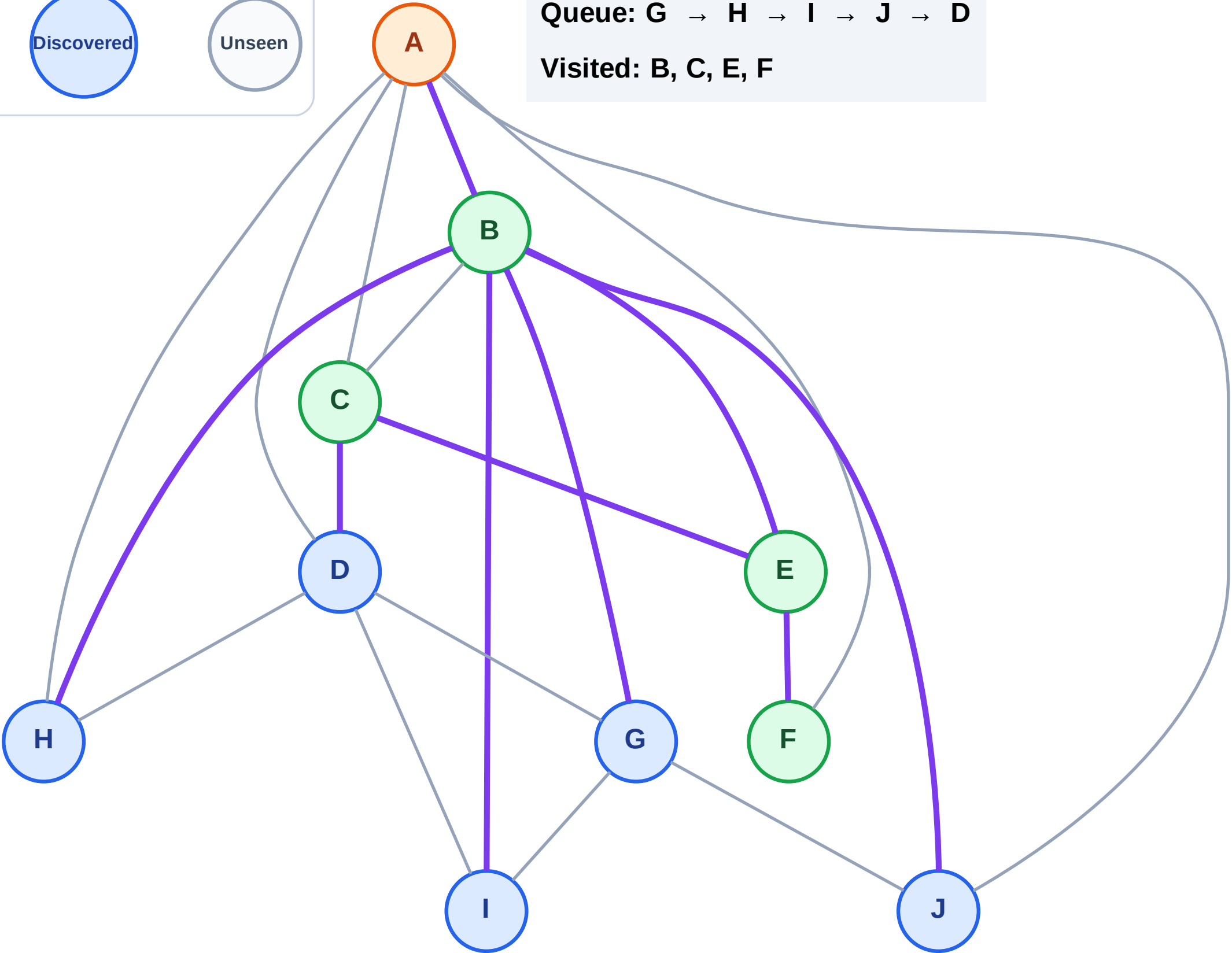
Processing

Discovered

Unseen

Queue: G → H → I → J → D

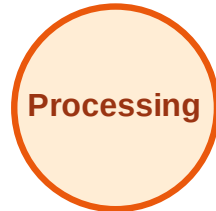
Visited: B, C, E, F



BFS Traversal

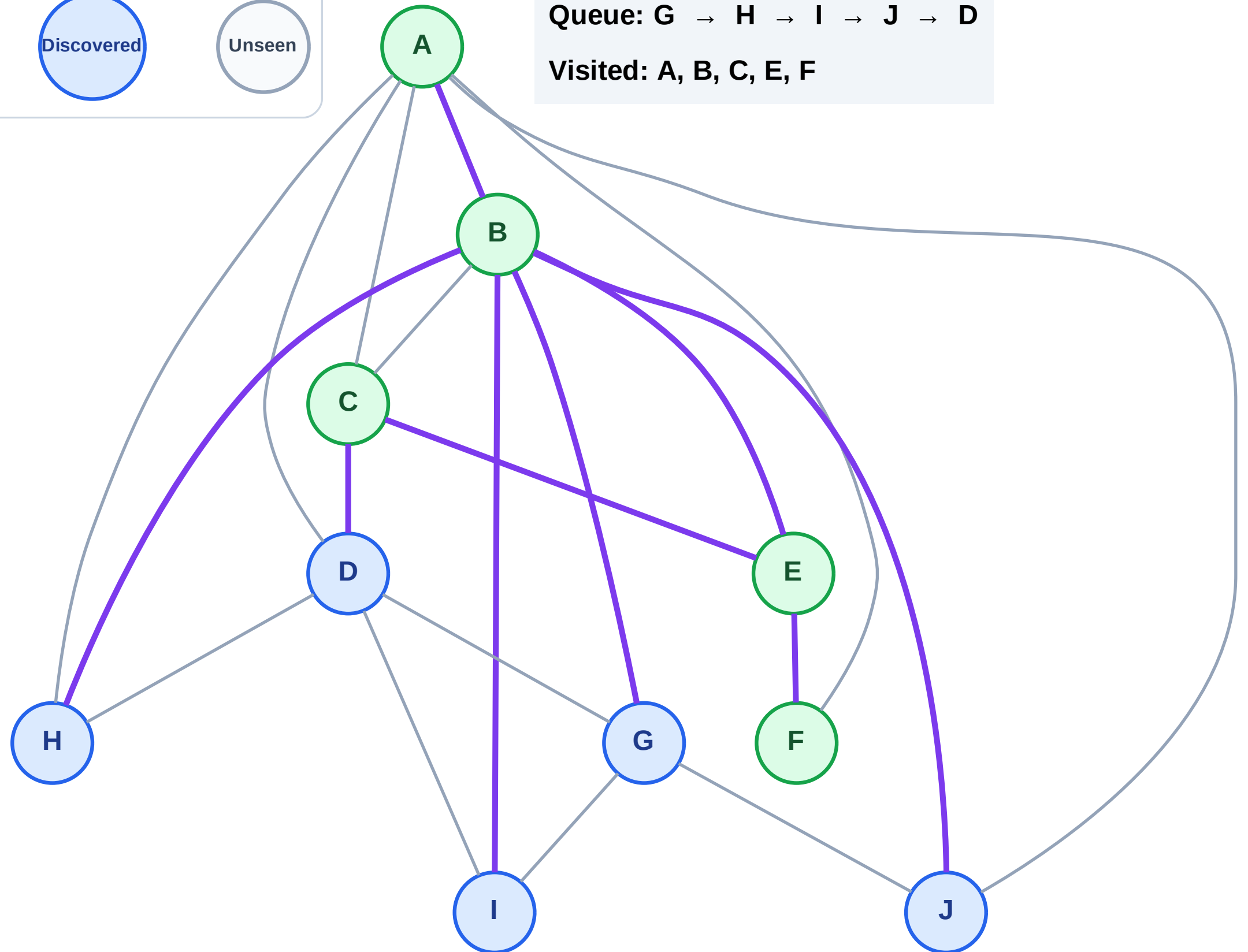
Step 11: No new neighbors from A

Legend



Queue: G → H → I → J → D

Visited: A, B, C, E, F



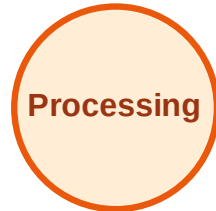
BFS Traversal

Step 12: Process node G

Legend



Complete



Processing



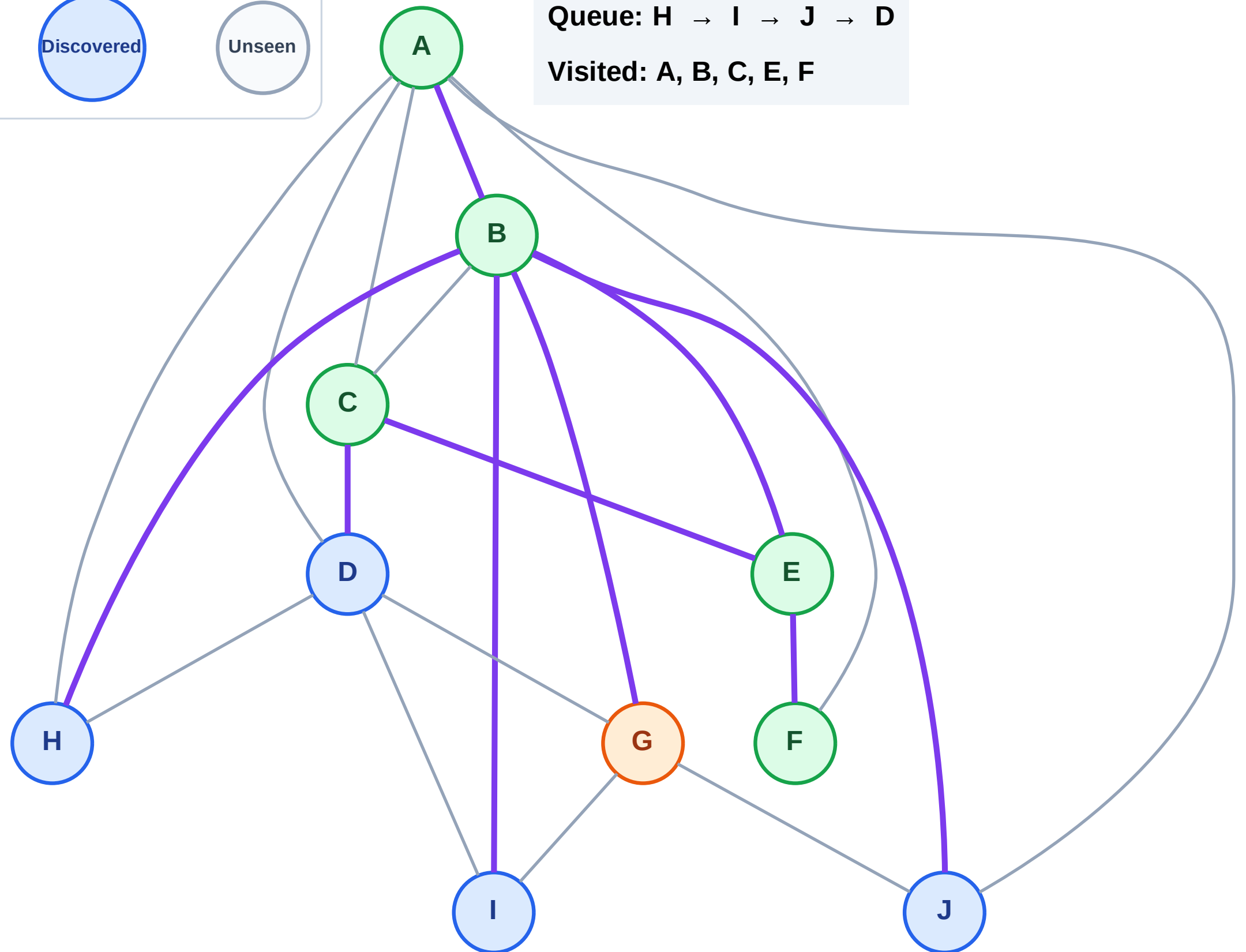
Discovered



Unseen

Queue: H → I → J → D

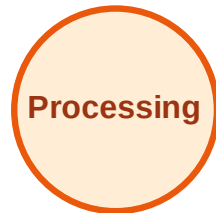
Visited: A, B, C, E, F



BFS Traversal

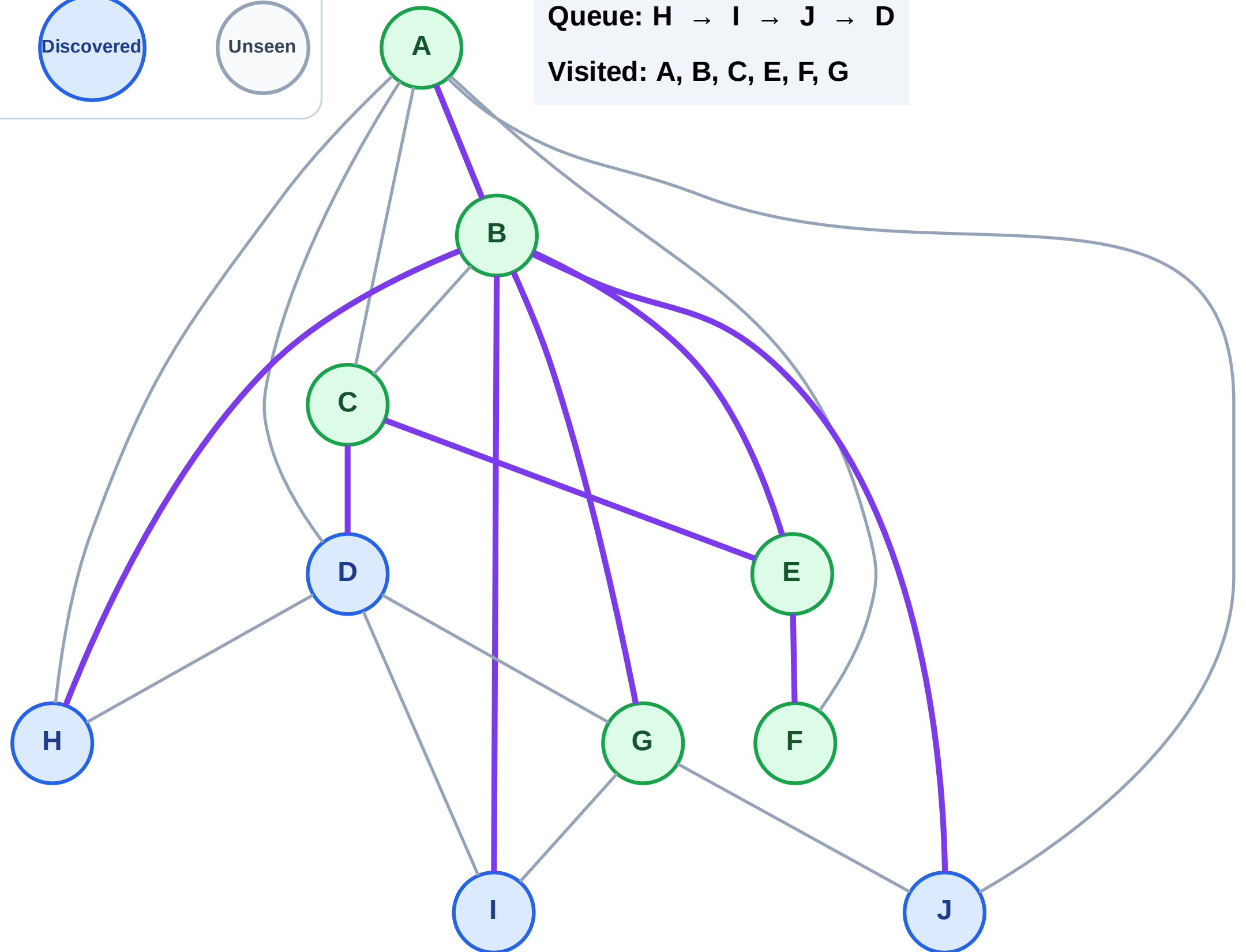
Step 13: No new neighbors from G

Legend



Queue: H → I → J → D

Visited: A, B, C, E, F, G



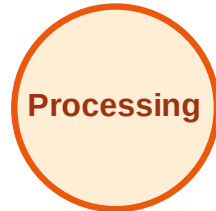
BFS Traversal

Step 14: Process node H

Legend



Complete



Processing



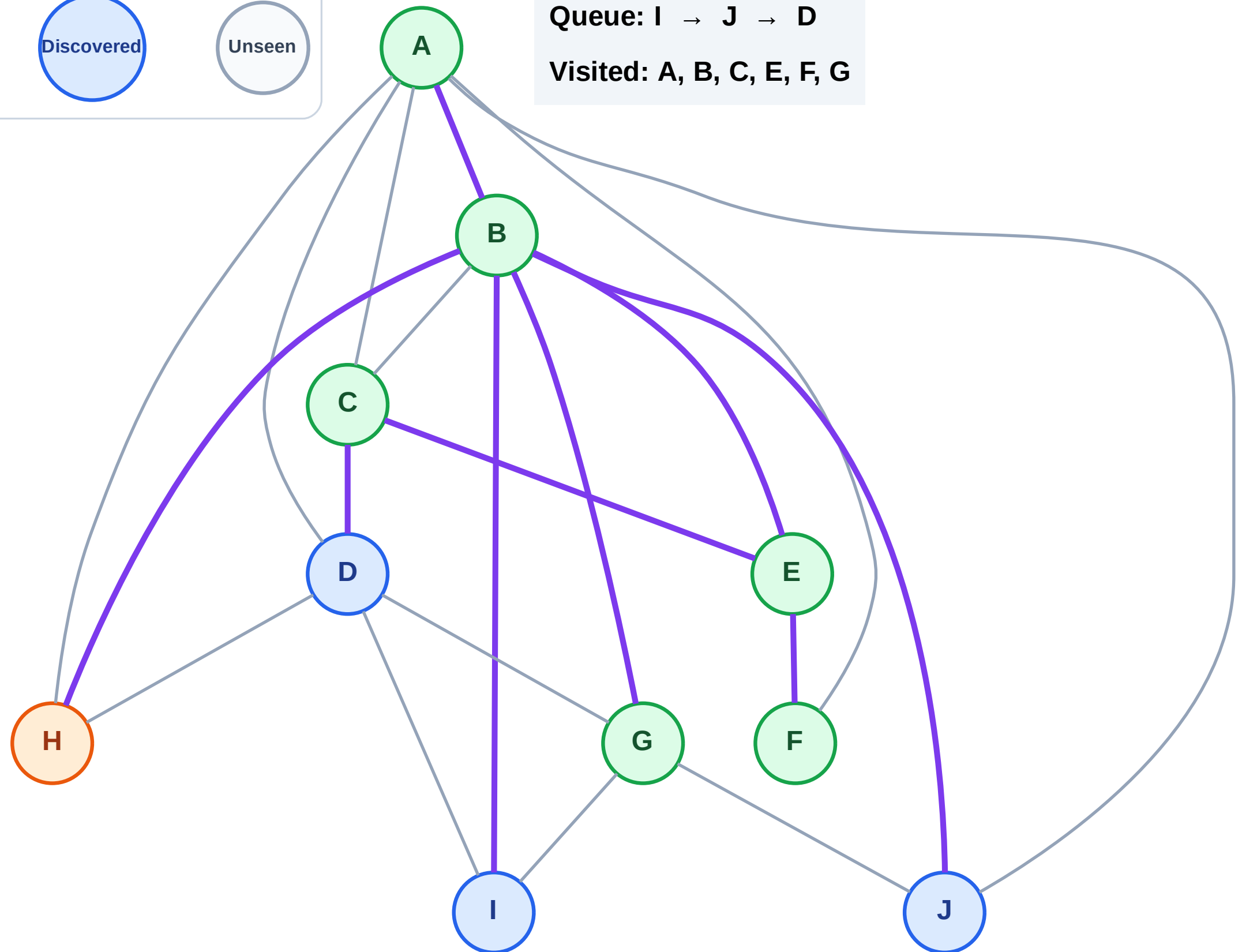
Discovered



Unseen

Queue: I → J → D

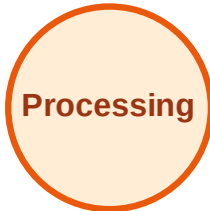
Visited: A, B, C, E, F, G



BFS Traversal

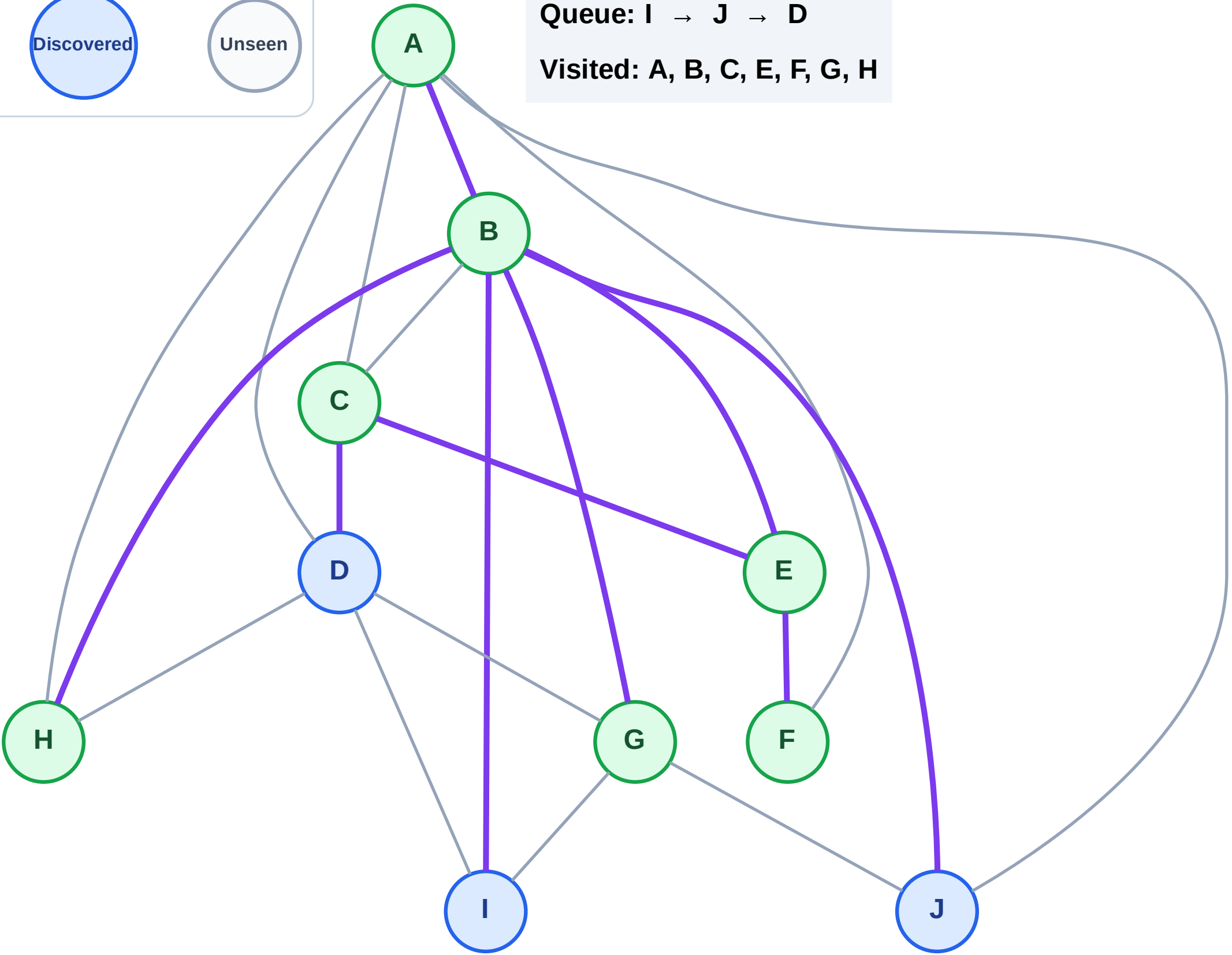
Step 15: No new neighbors from H

Legend



Queue: I → J → D

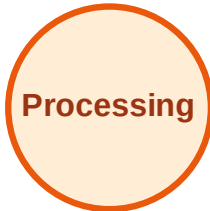
Visited: A, B, C, E, F, G, H



BFS Traversal

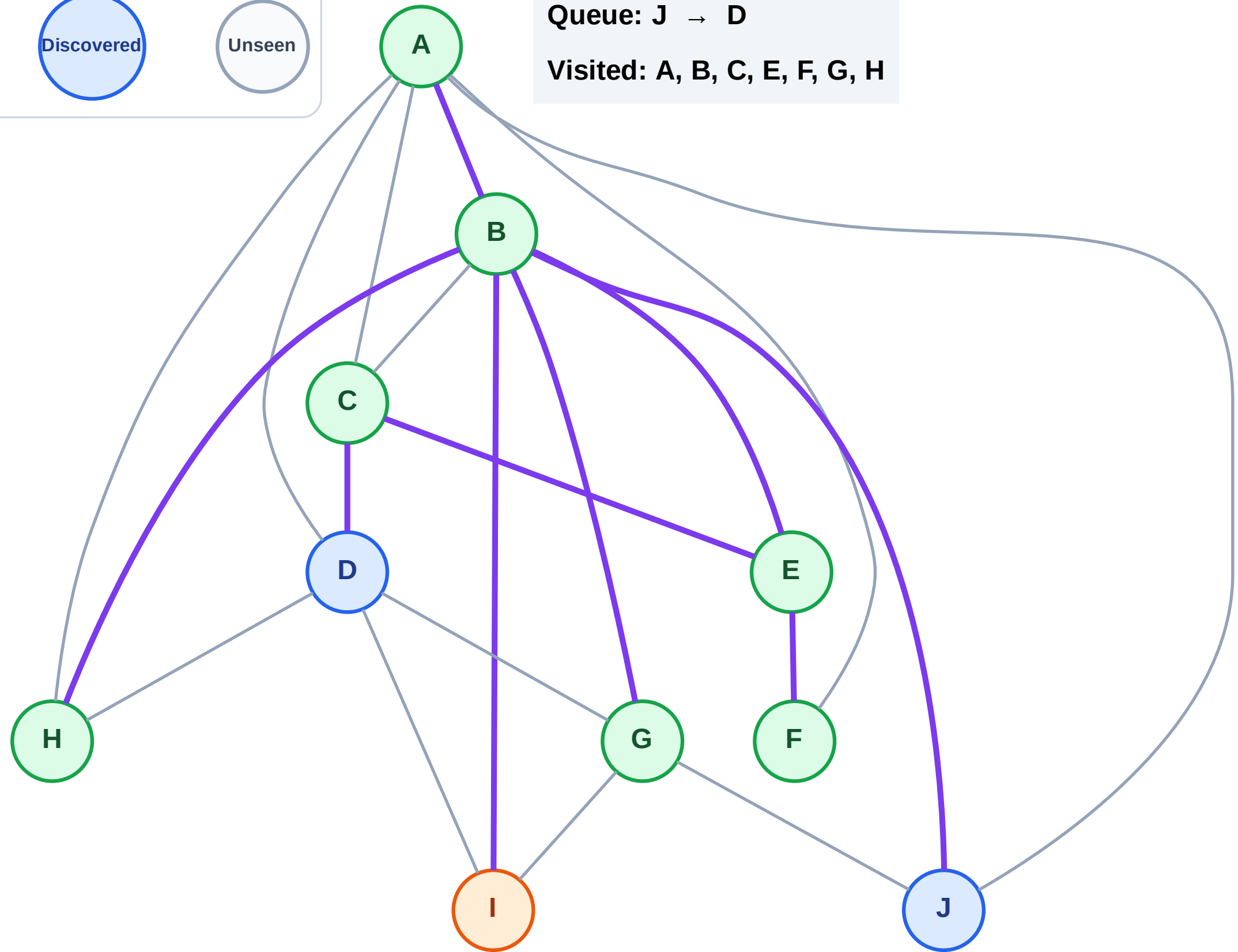
Step 16: Process node I

Legend



Queue: J → D

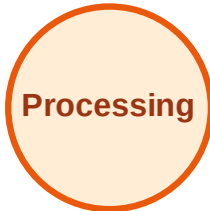
Visited: A, B, C, E, F, G, H



BFS Traversal

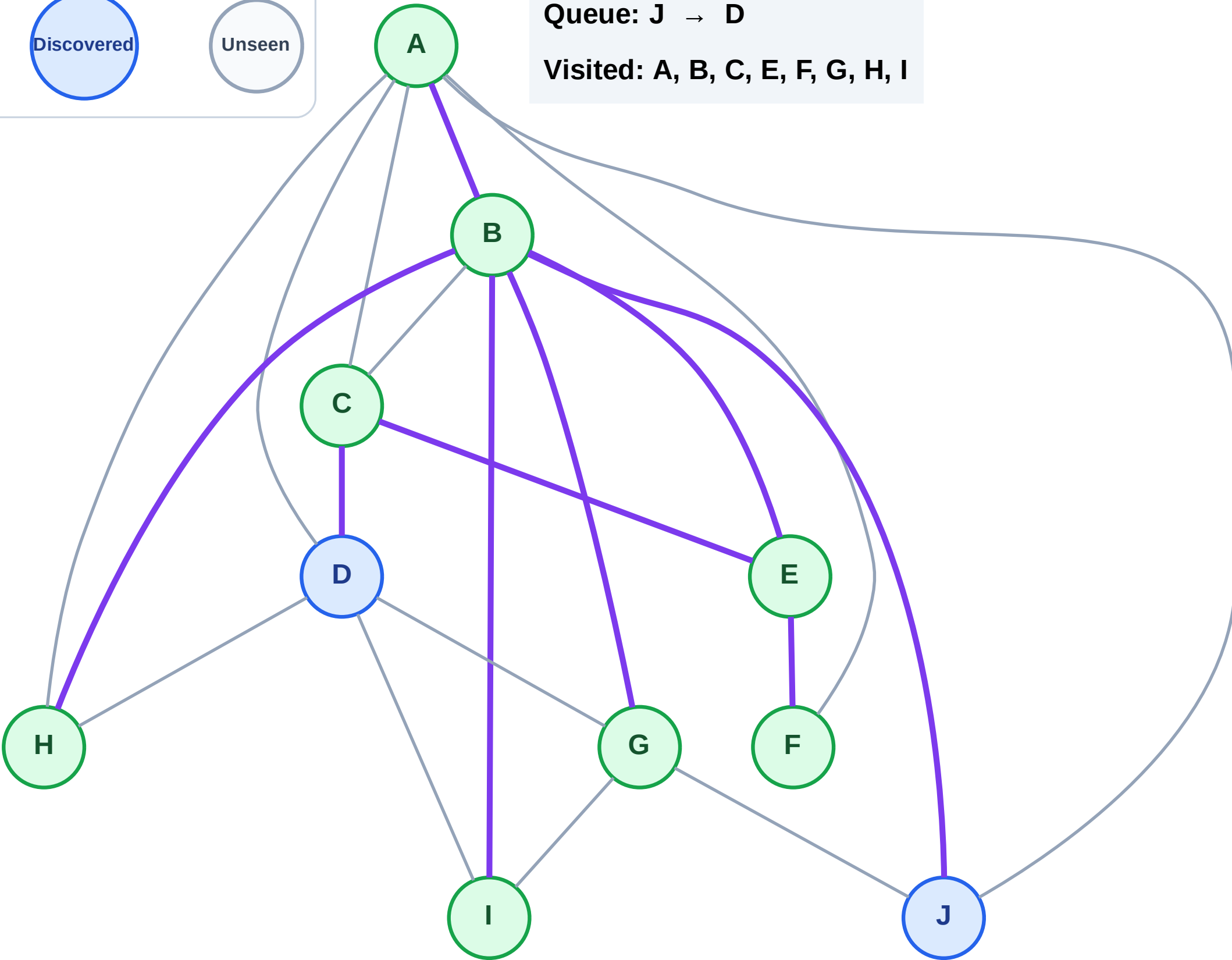
Step 17: No new neighbors from I

Legend



Queue: J → D

Visited: A, B, C, E, F, G, H, I



BFS Traversal

Step 18: Process node J

Legend

Complete

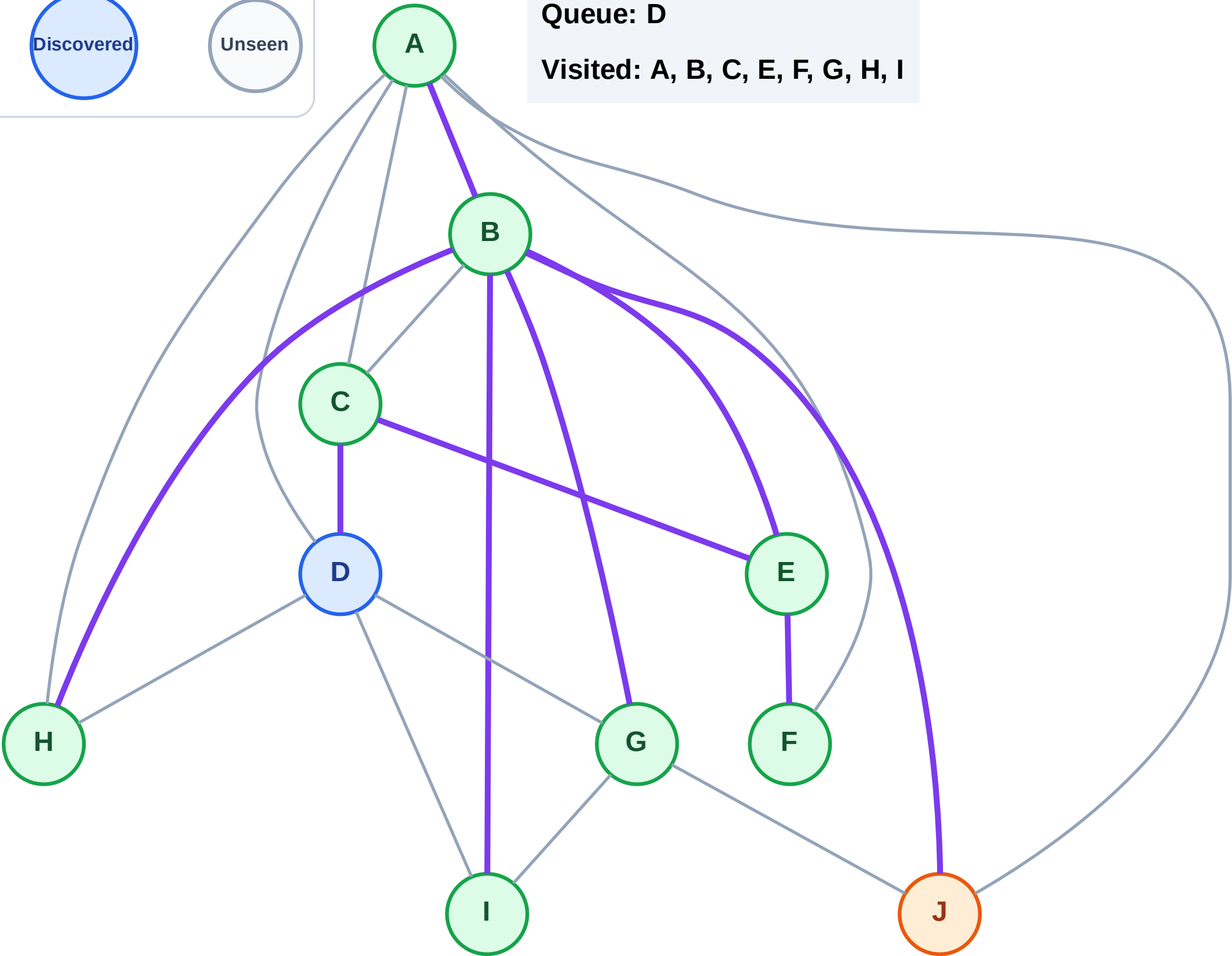
Processing

Discovered

Unseen

Queue: D

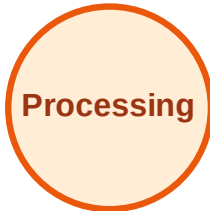
Visited: A, B, C, E, F, G, H, I



BFS Traversal

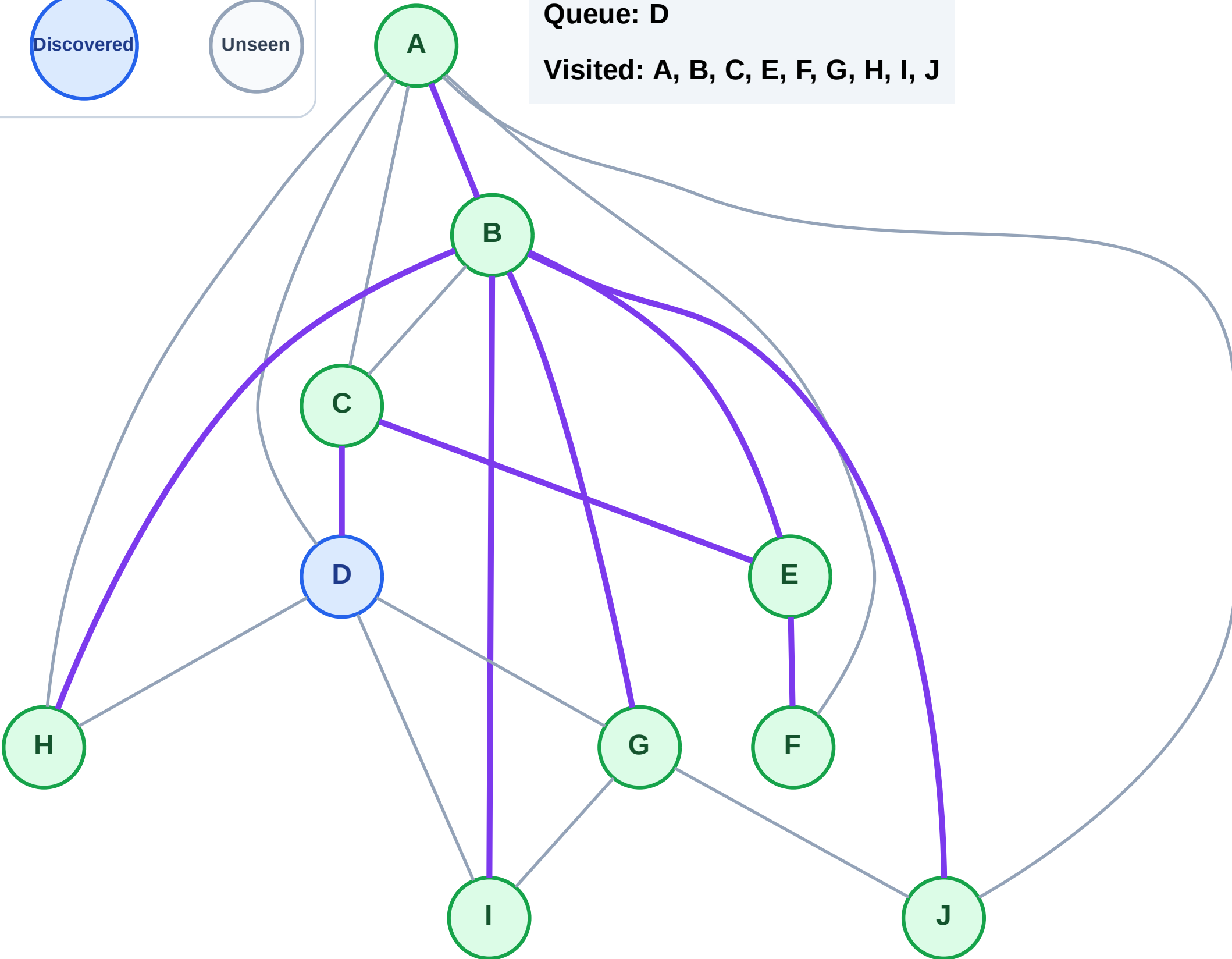
Step 19: No new neighbors from J

Legend



Queue: D

Visited: A, B, C, E, F, G, H, I, J



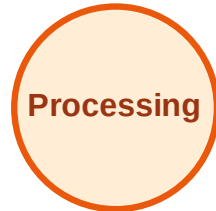
BFS Traversal

Step 20: Process node D

Legend



Complete



Processing



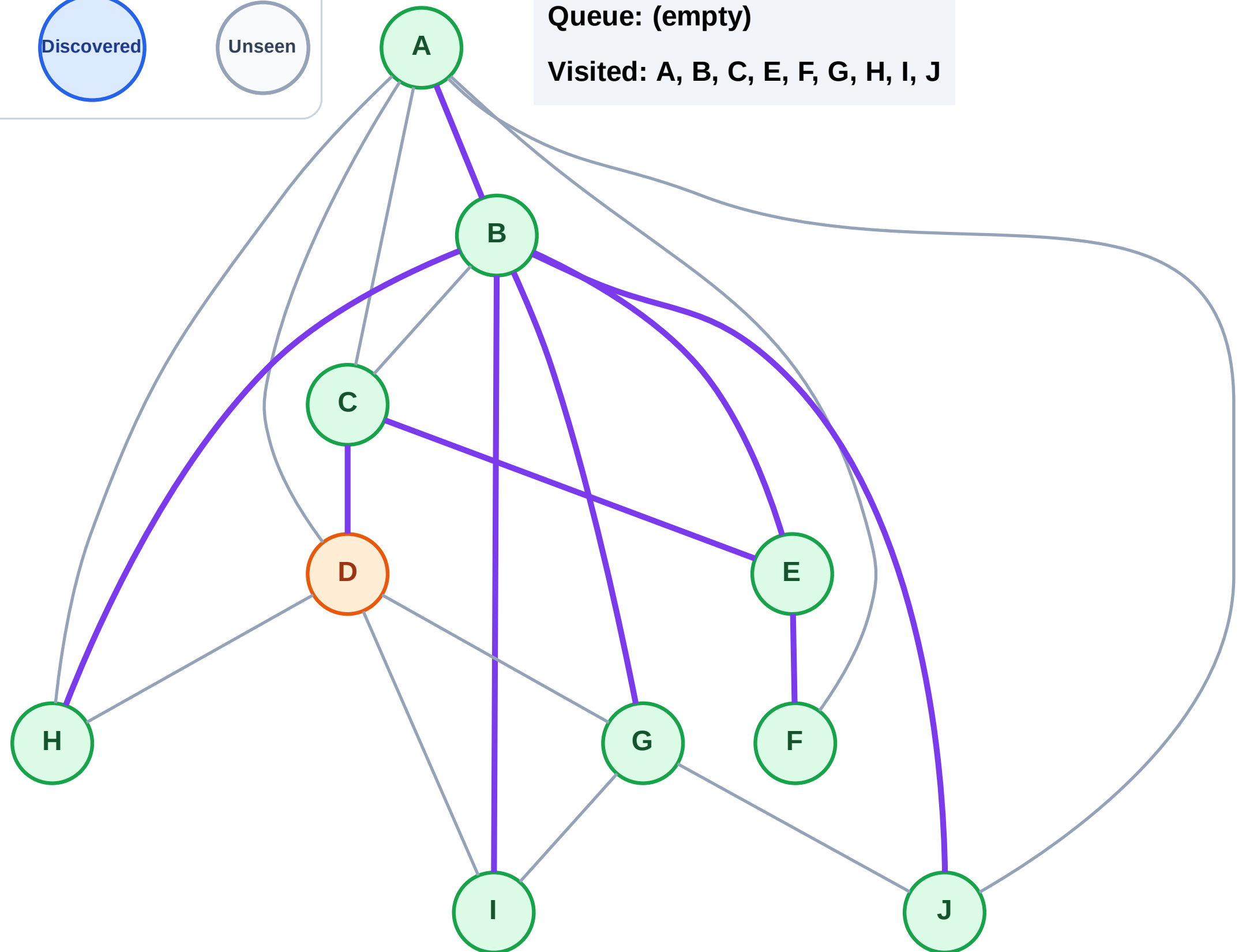
Discovered



Unseen

Queue: (empty)

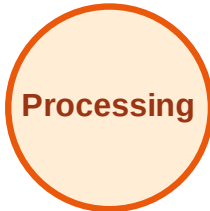
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

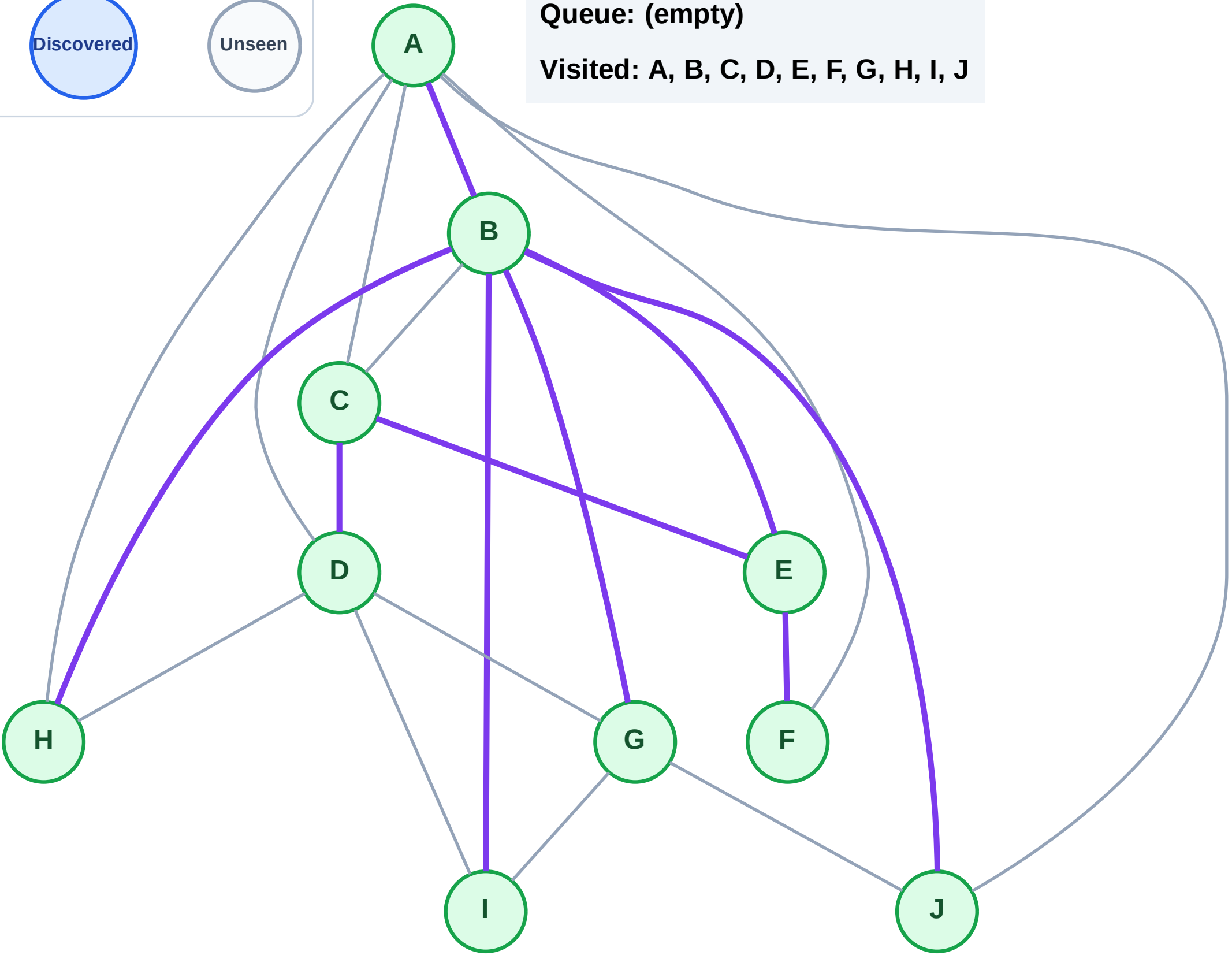
Step 21: No new neighbors from D

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

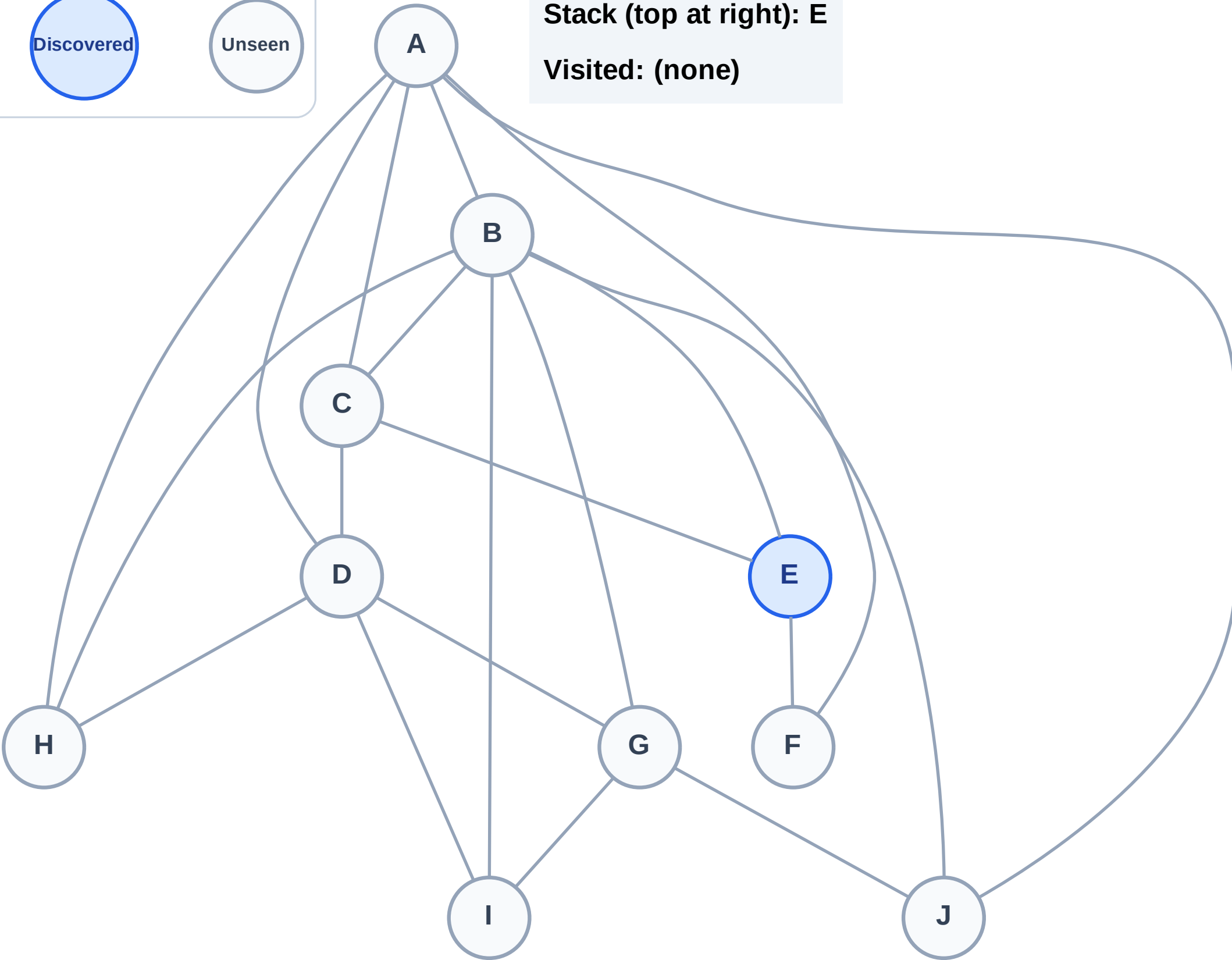
Step 1: Start at E

Legend



Stack (top at right): E

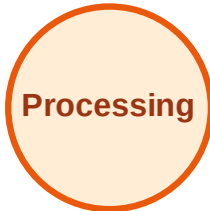
Visited: (none)



DFS Traversal

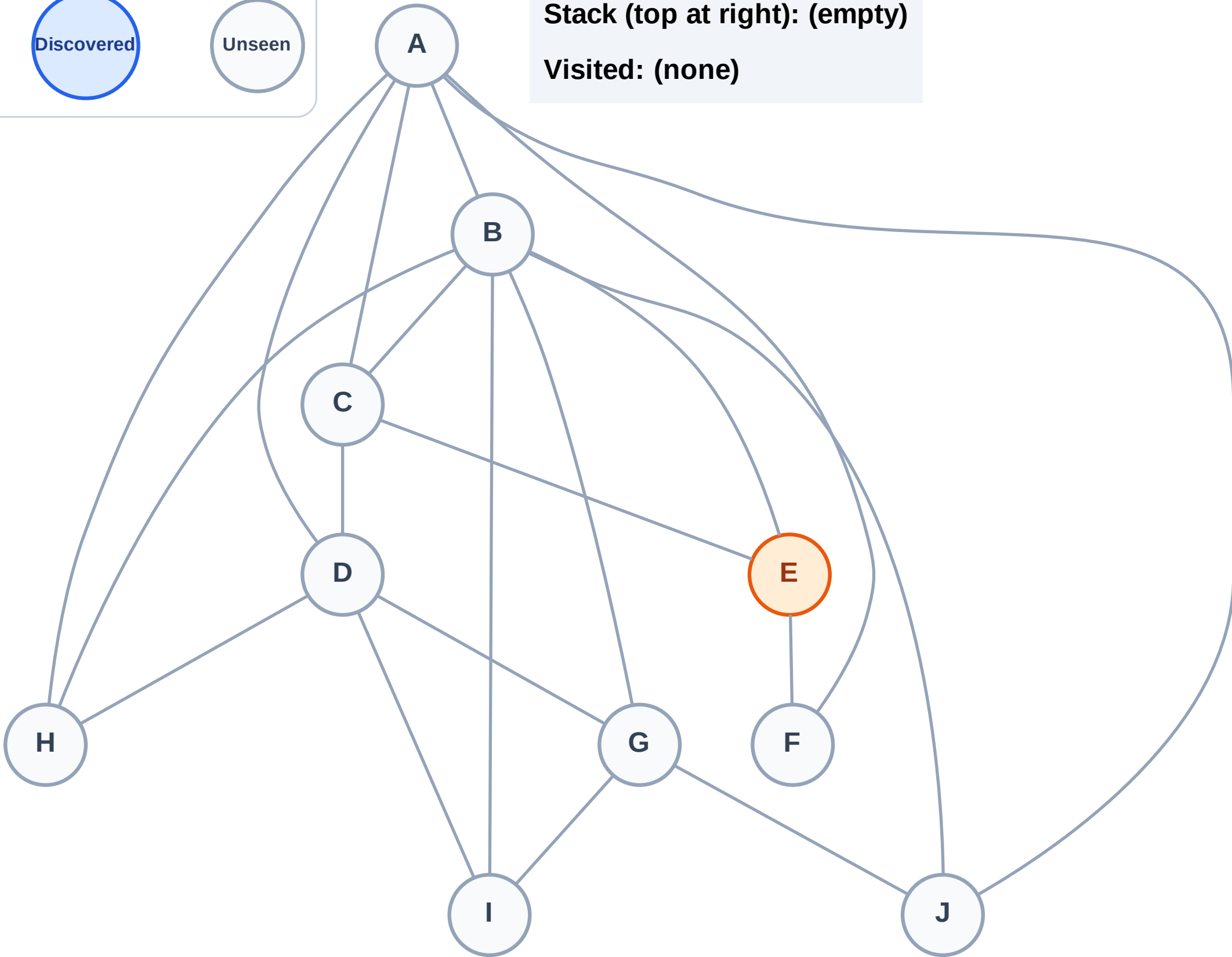
Step 2: Process node E

Legend



Stack (top at right): (empty)

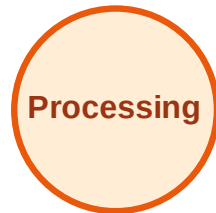
Visited: (none)



DFS Traversal

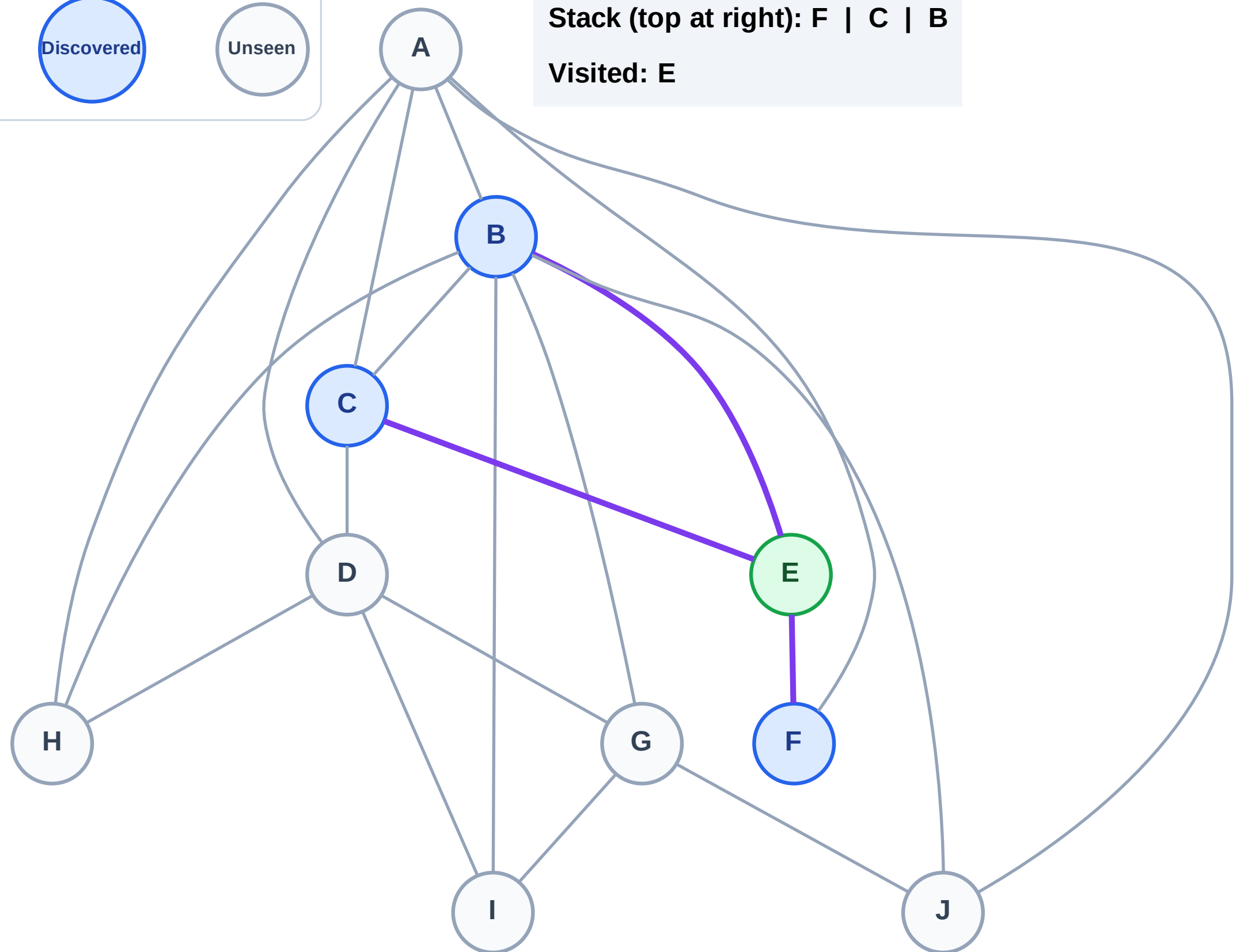
Step 3: Discover F, C, B from E

Legend



Stack (top at right): F | C | B

Visited: E



DFS Traversal

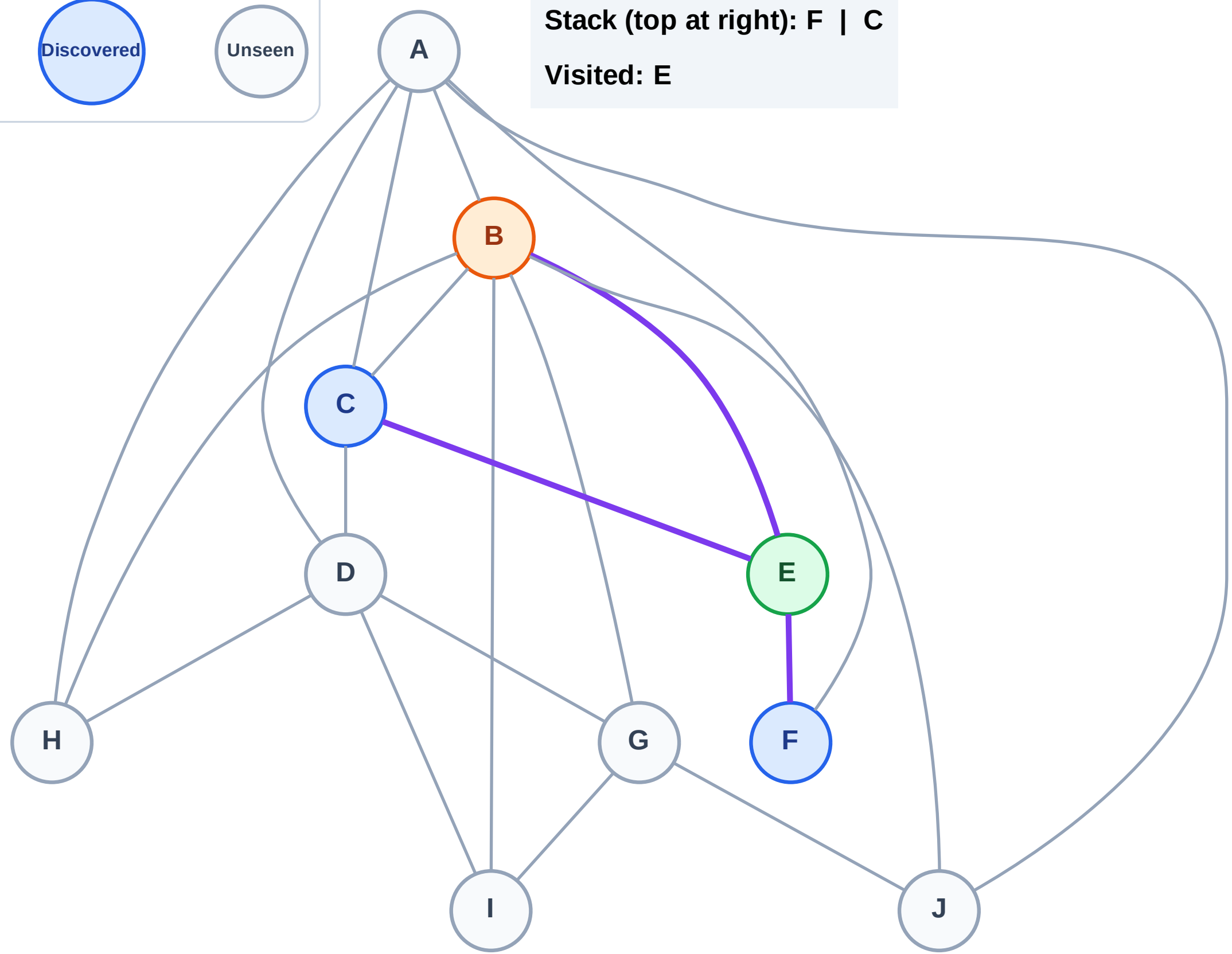
Step 4: Process node B

Legend



Stack (top at right): F | C

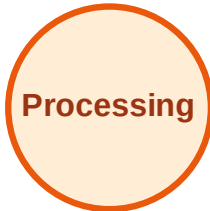
Visited: E



DFS Traversal

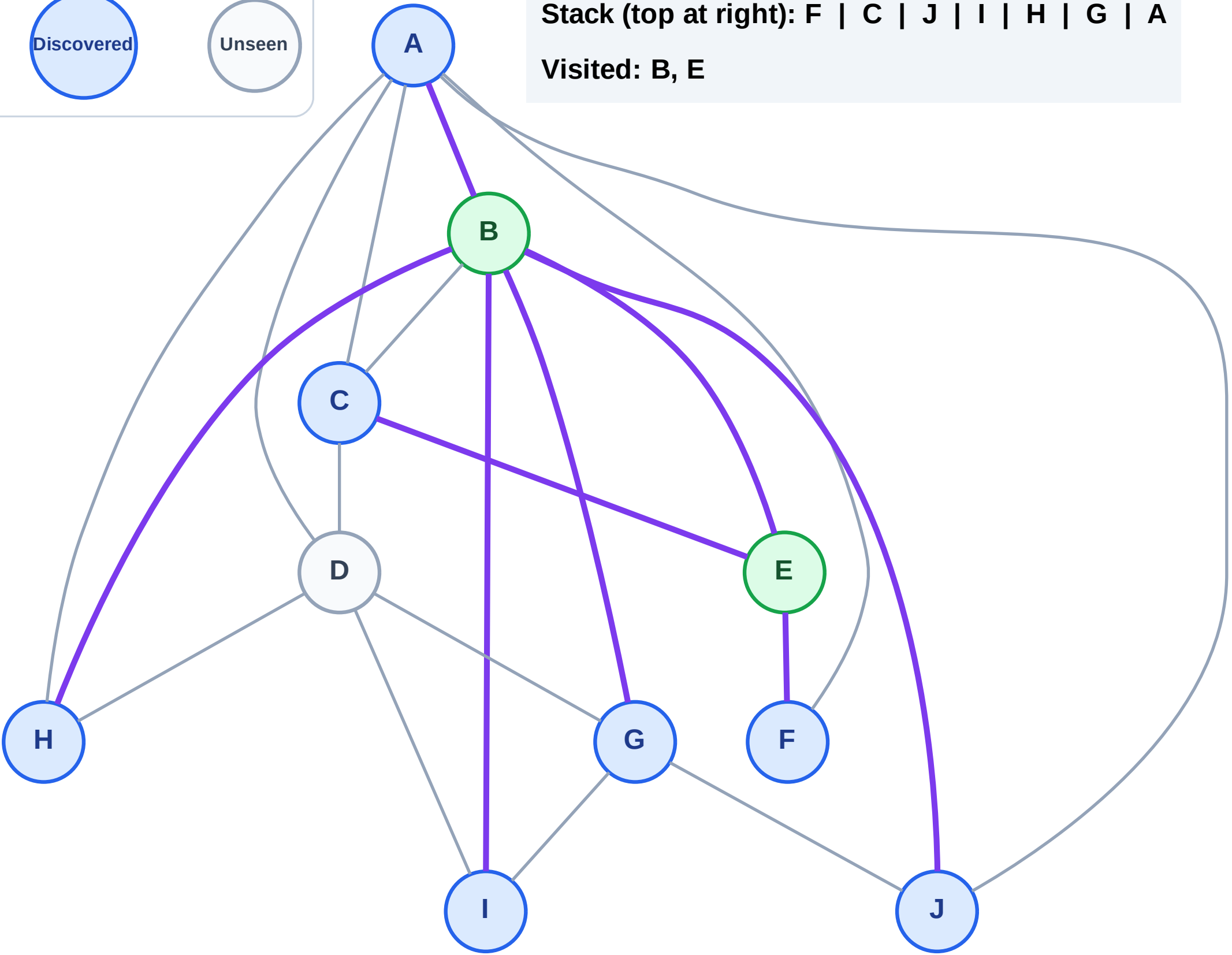
Step 5: Discover J, I, H, G, A from B

Legend



Stack (top at right): F | C | J | I | H | G | A

Visited: B, E



DFS Traversal

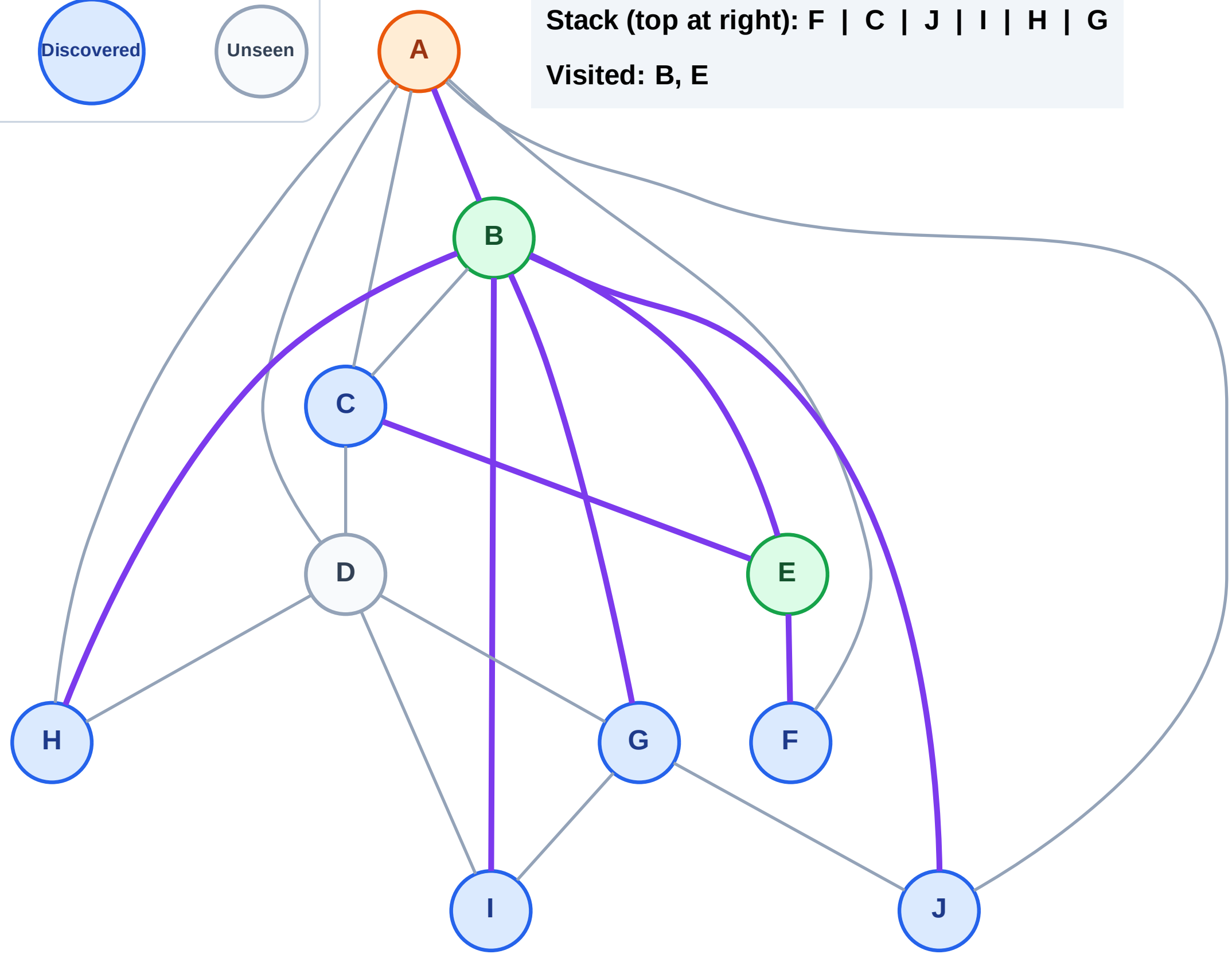
Step 6: Process node A

Legend



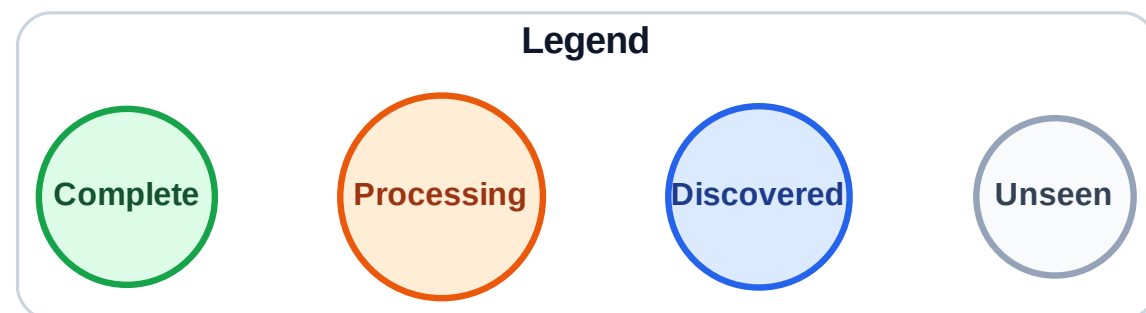
Stack (top at right): F | C | J | I | H | G

Visited: B, E

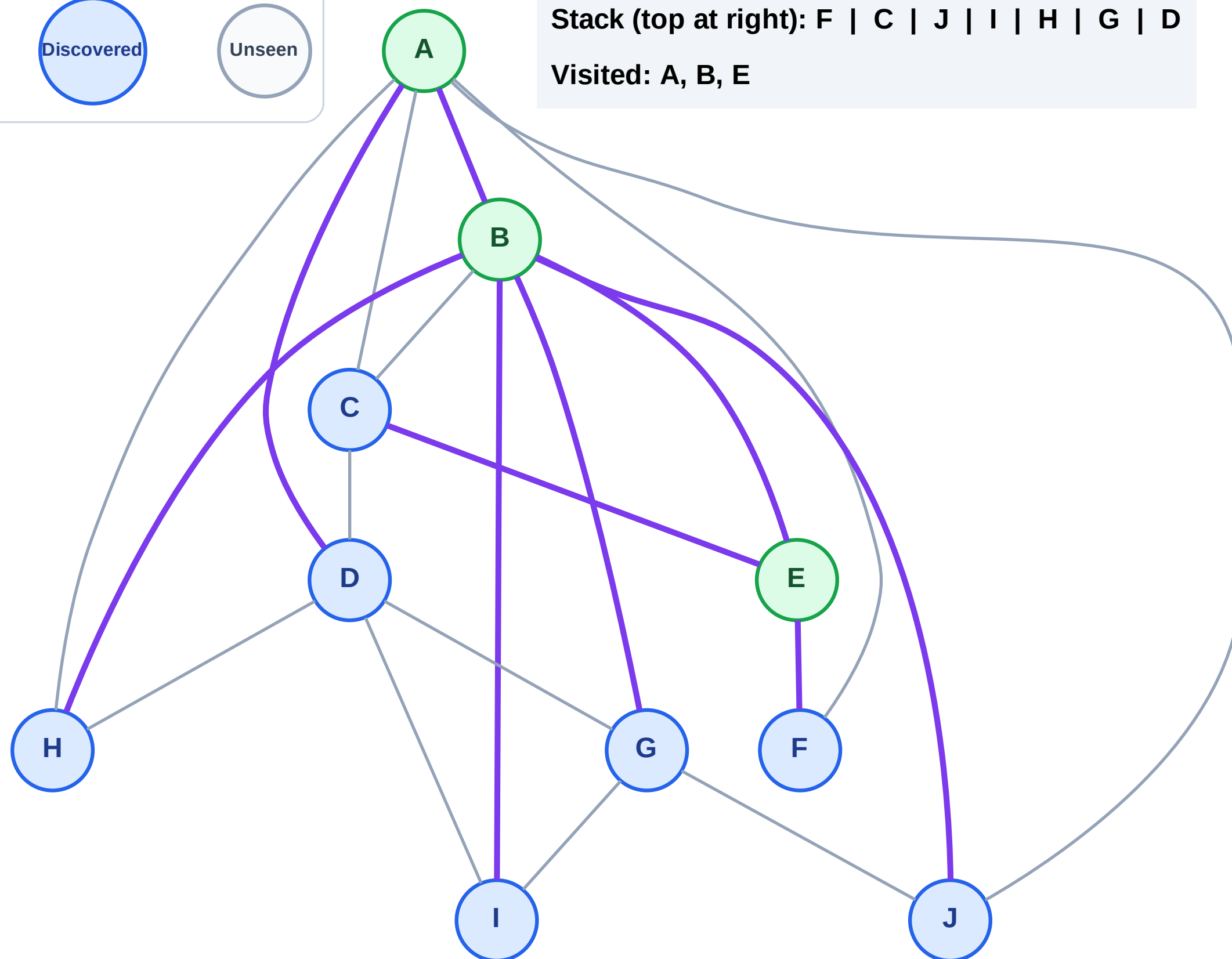


DFS Traversal

Step 7: Discover D from A



Stack (top at right): F | C | J | I | H | G | D
Visited: A, B, E



DFS Traversal

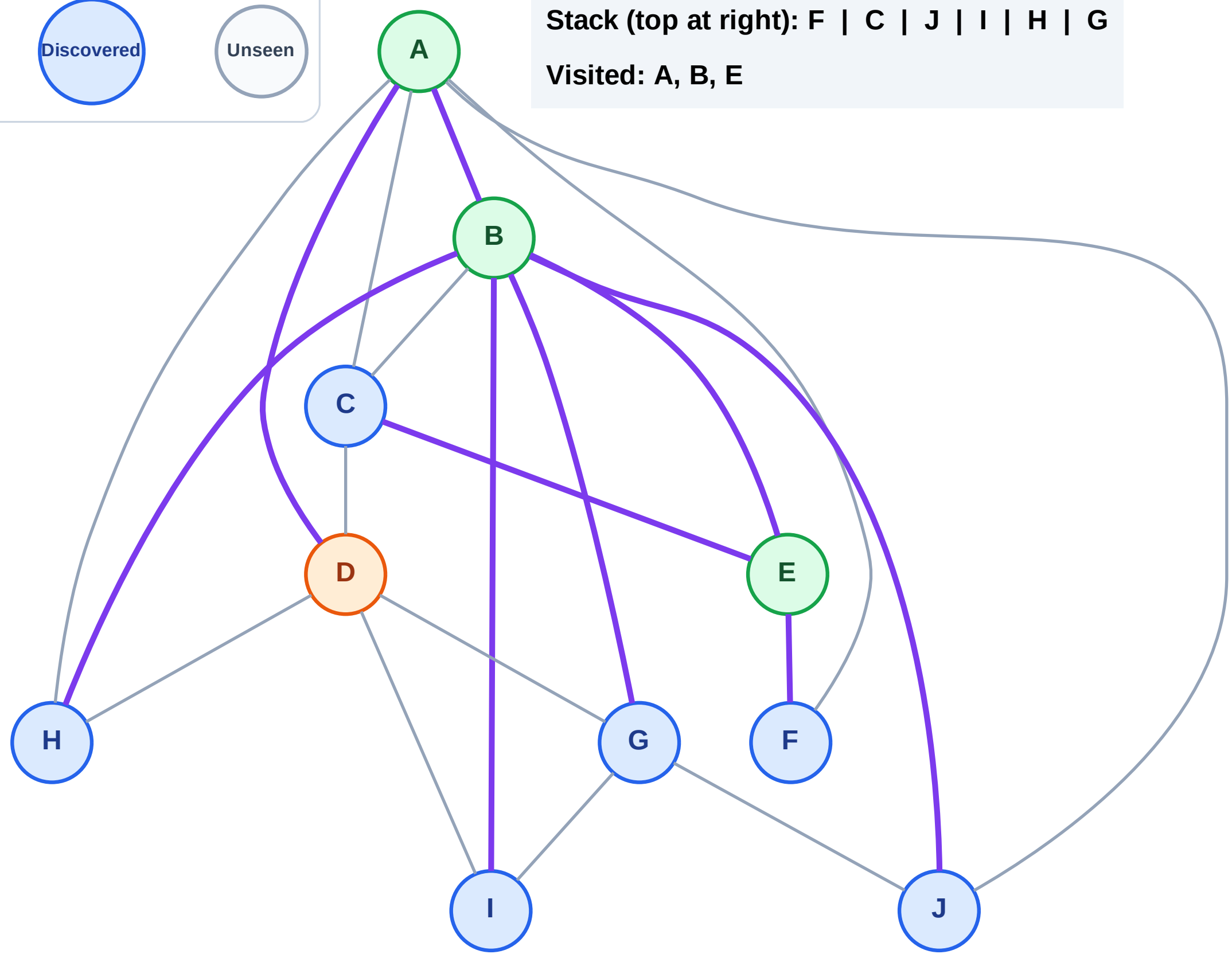
Step 8: Process node D

Legend



Stack (top at right): F | C | J | I | H | G

Visited: A, B, E



DFS Traversal

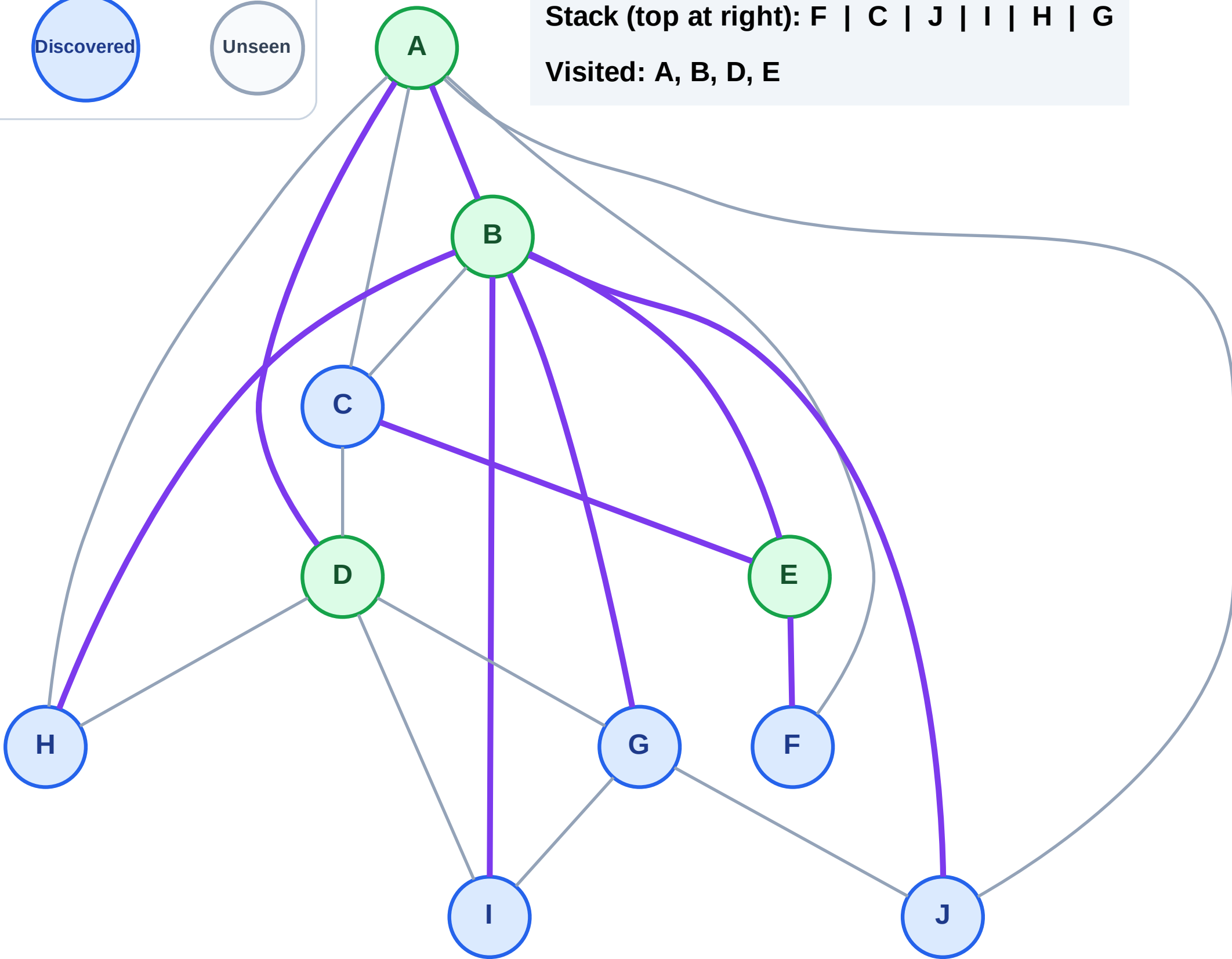
Step 9: No new neighbors from D

Legend



Stack (top at right): F | C | J | I | H | G

Visited: A, B, D, E



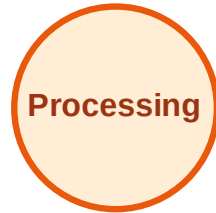
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



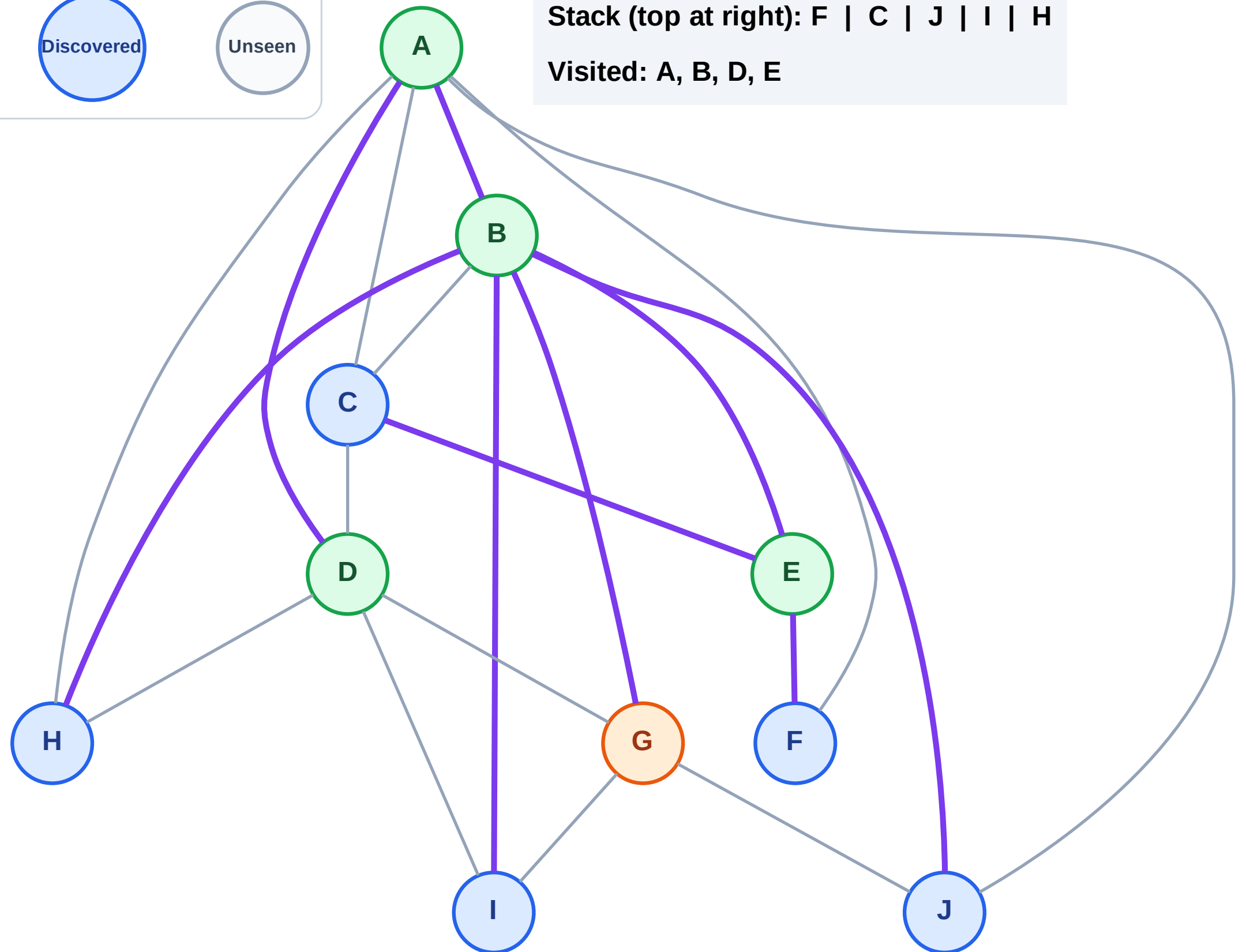
Discovered



Unseen

Stack (top at right): F | C | J | I | H

Visited: A, B, D, E



DFS Traversal

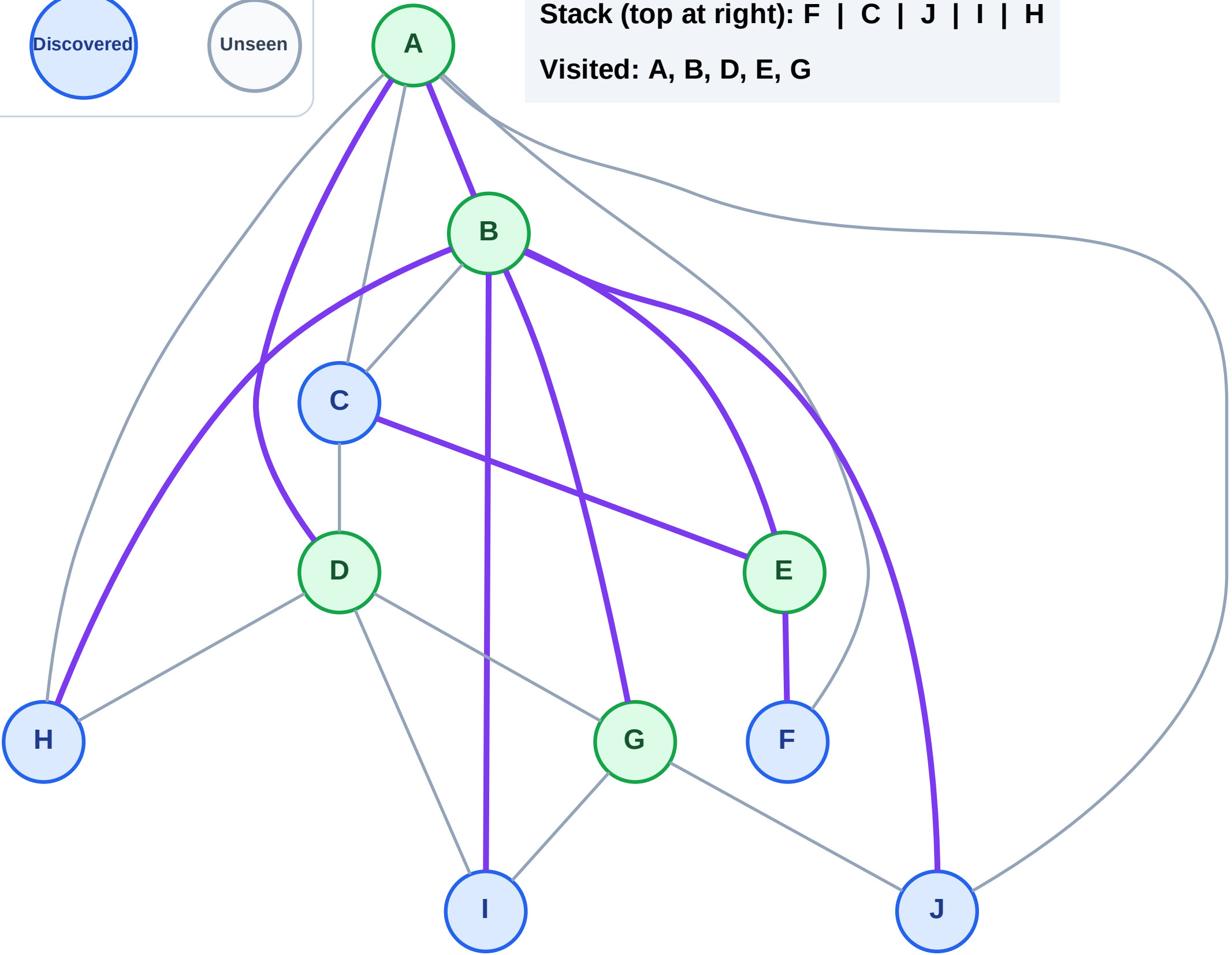
Step 11: No new neighbors from G

Legend



Stack (top at right): F | C | J | I | H

Visited: A, B, D, E, G



DFS Traversal

Step 12: Process node H

Legend

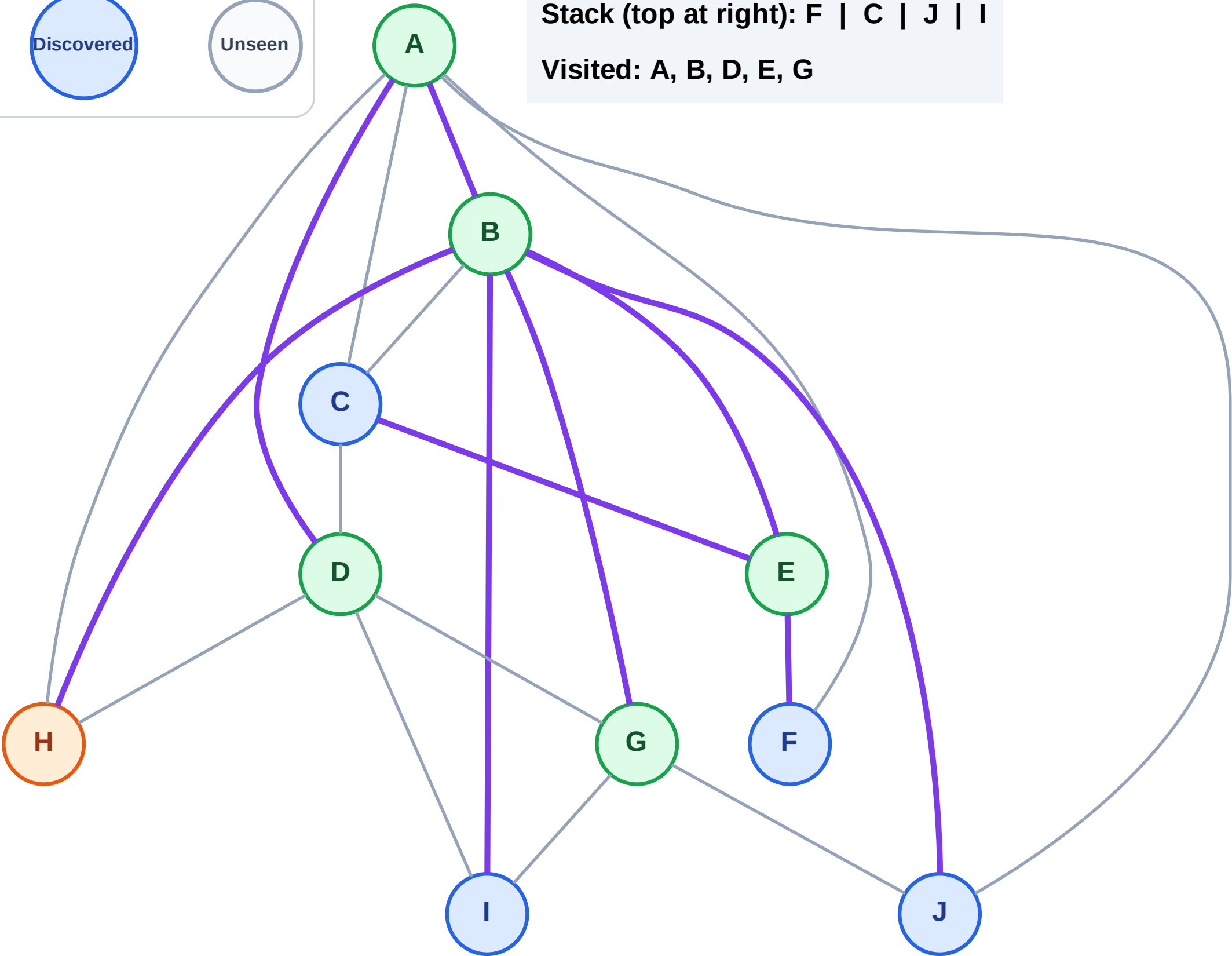
Complete

Processing

Discovered

Unseen

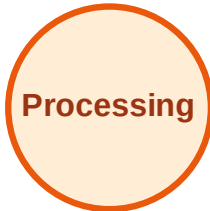
Stack (top at right): F | C | J | I
Visited: A, B, D, E, G



DFS Traversal

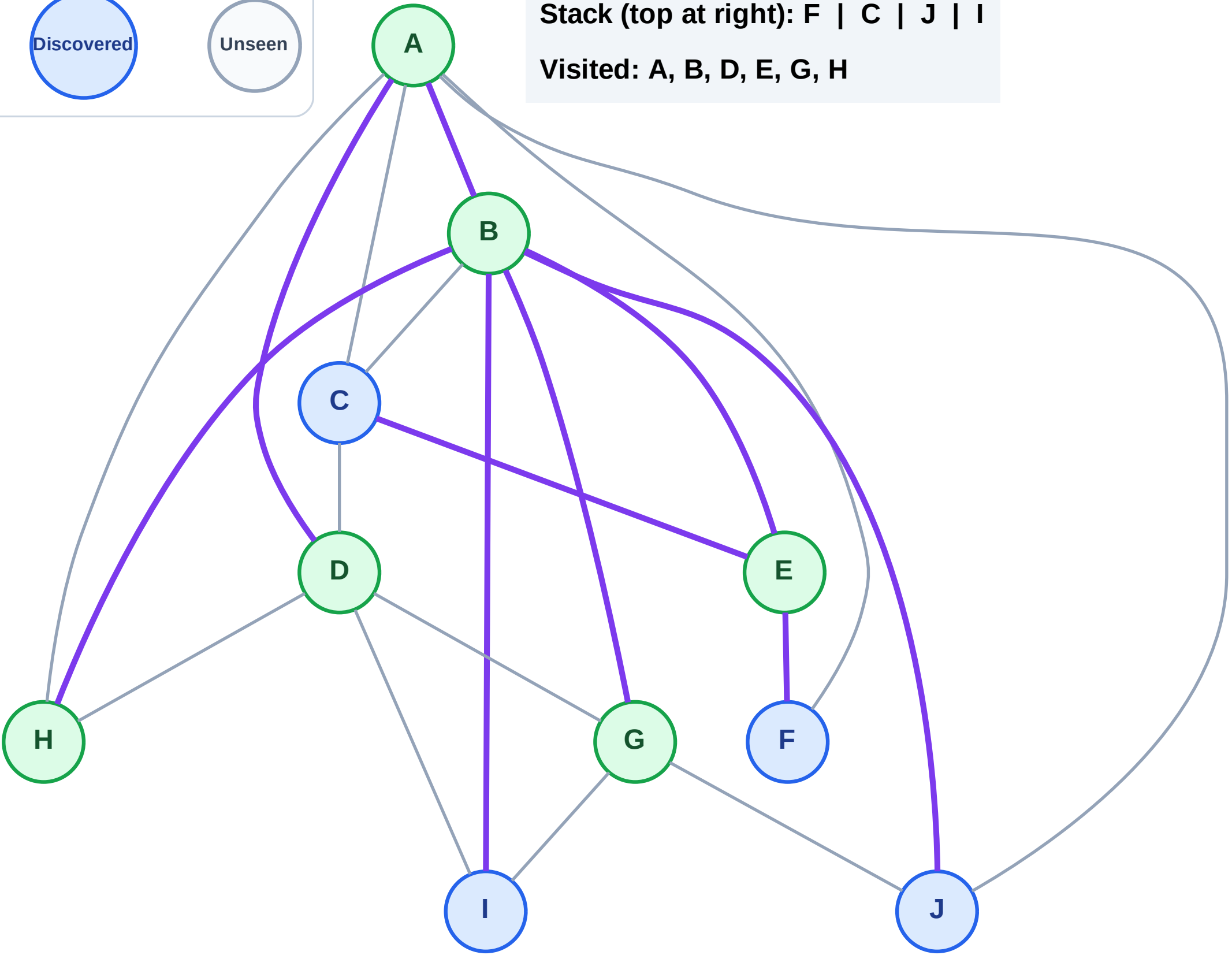
Step 13: No new neighbors from H

Legend



Stack (top at right): F | C | J | I

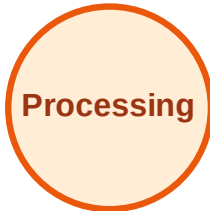
Visited: A, B, D, E, G, H



DFS Traversal

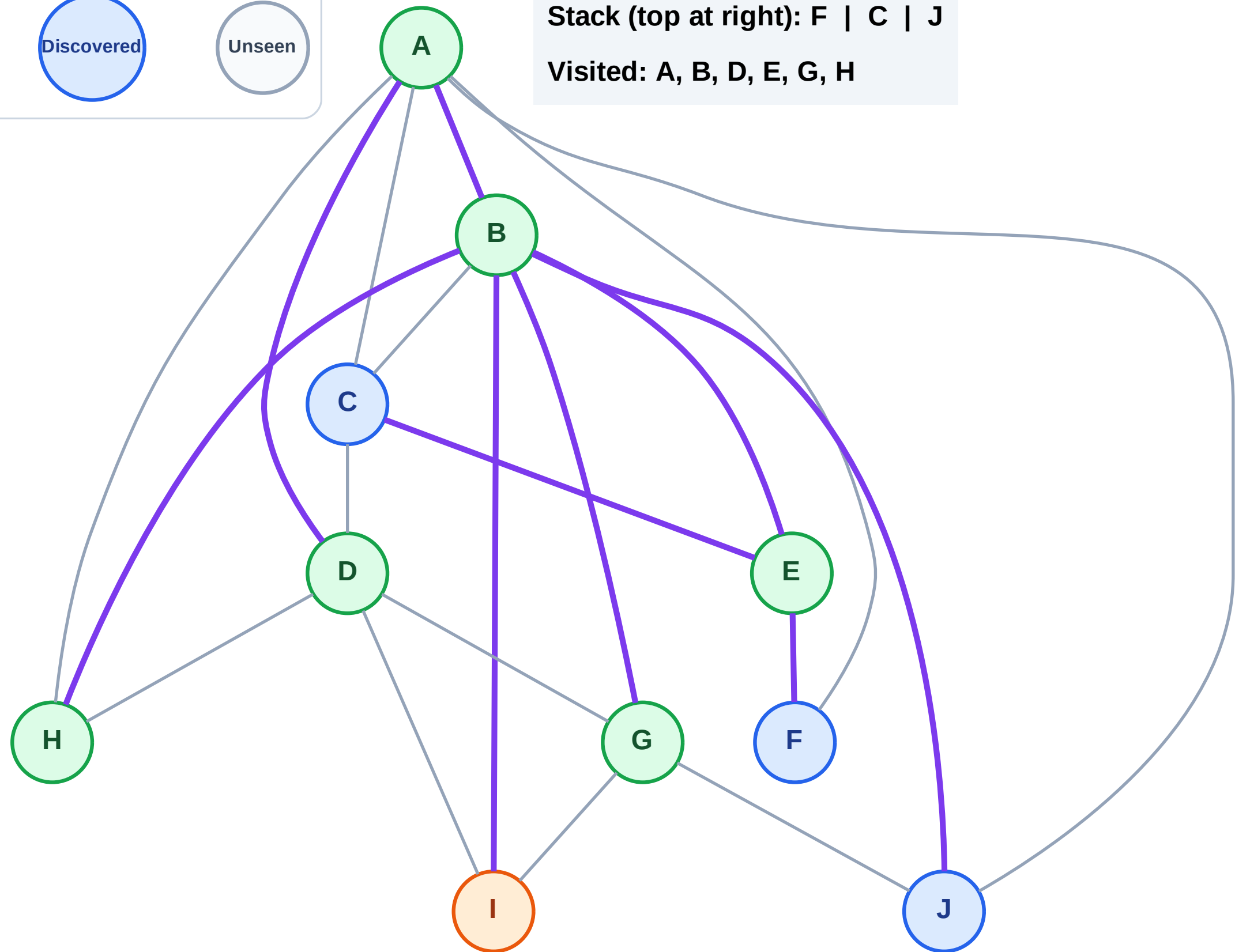
Step 14: Process node I

Legend



Stack (top at right): F | C | J

Visited: A, B, D, E, G, H



DFS Traversal

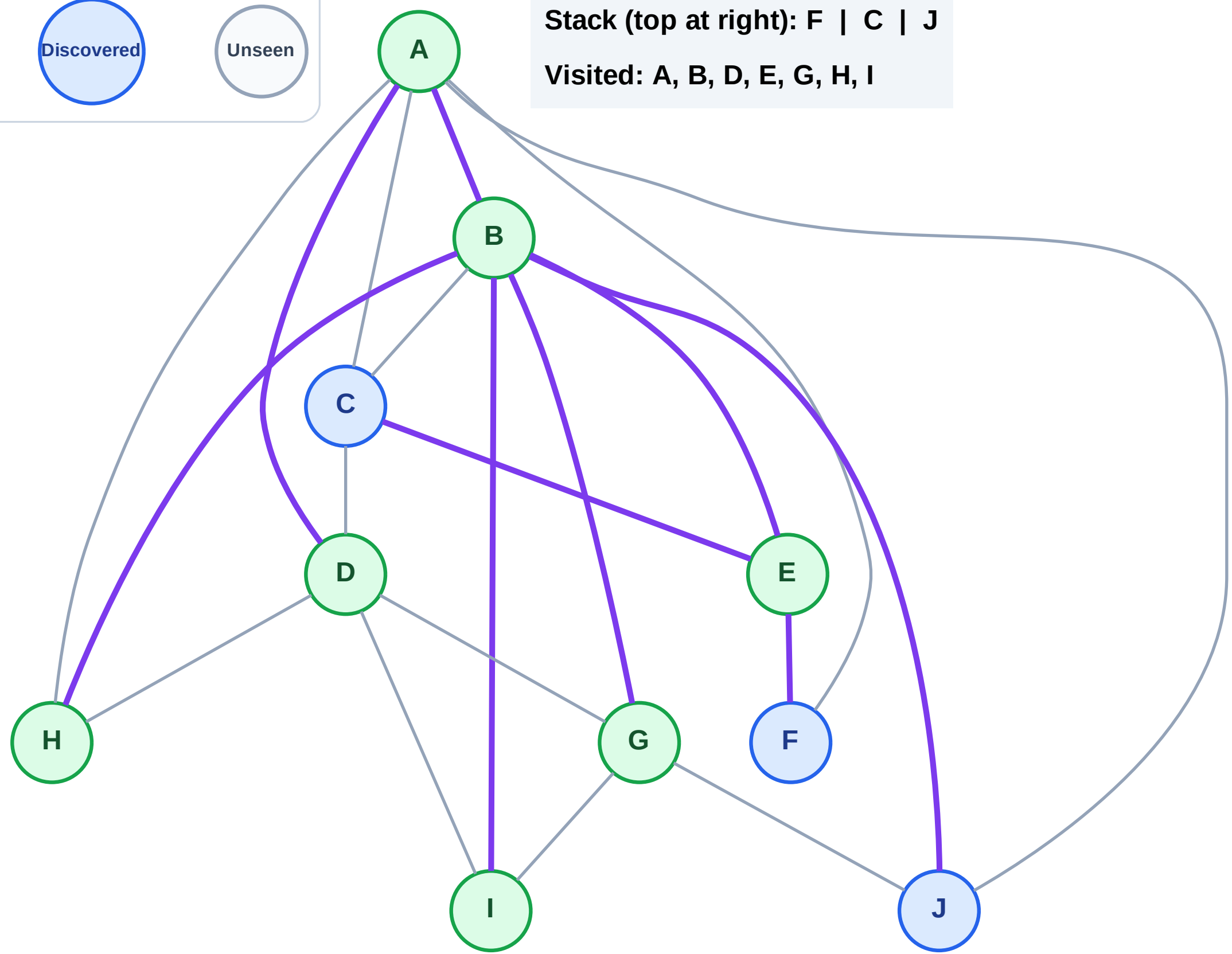
Step 15: No new neighbors from I

Legend



Stack (top at right): F | C | J

Visited: A, B, D, E, G, H, I



DFS Traversal

Step 16: Process node J

Legend

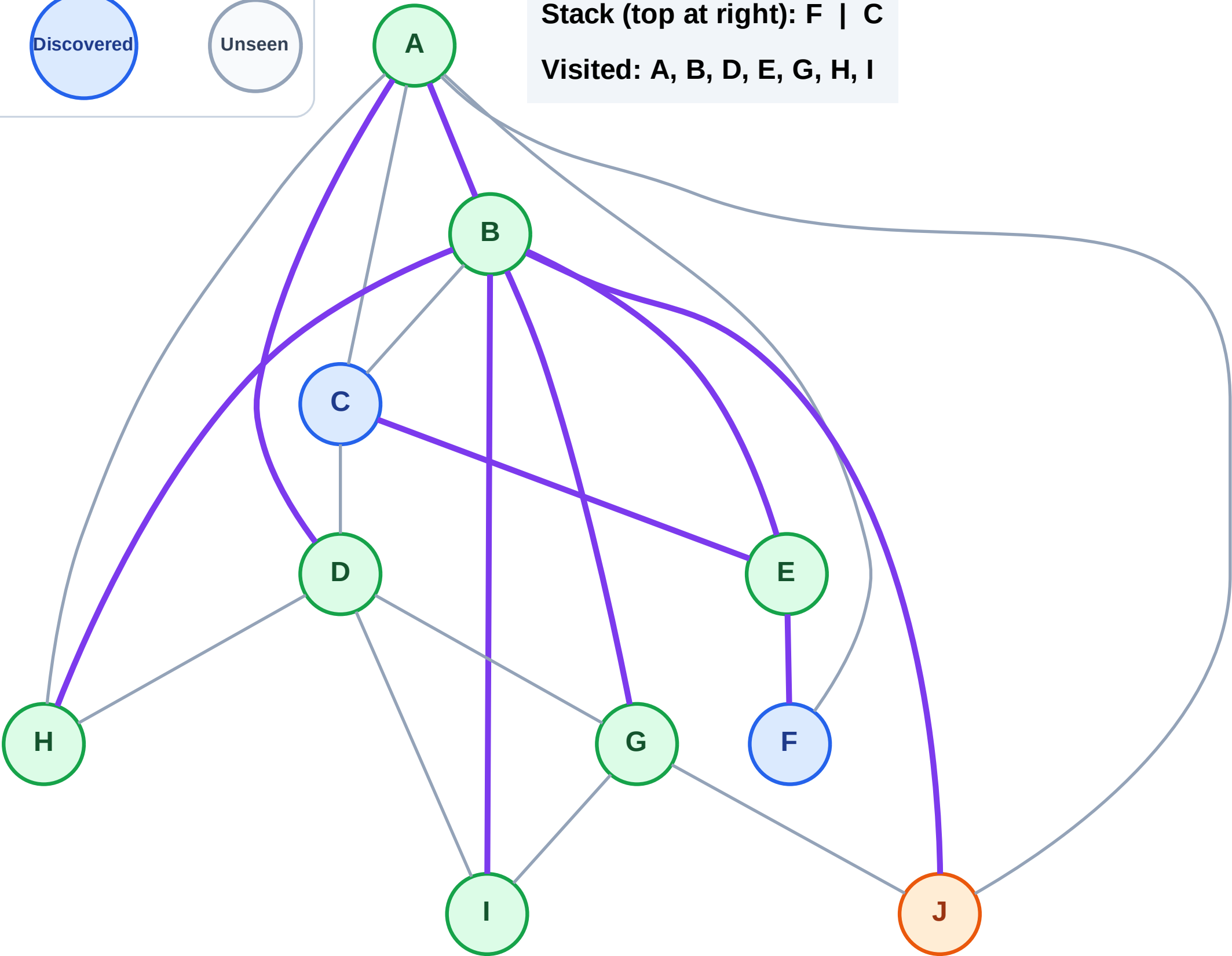
Complete

Processing

Discovered

Unseen

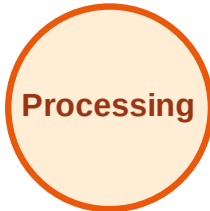
Stack (top at right): F | C
Visited: A, B, D, E, G, H, I



DFS Traversal

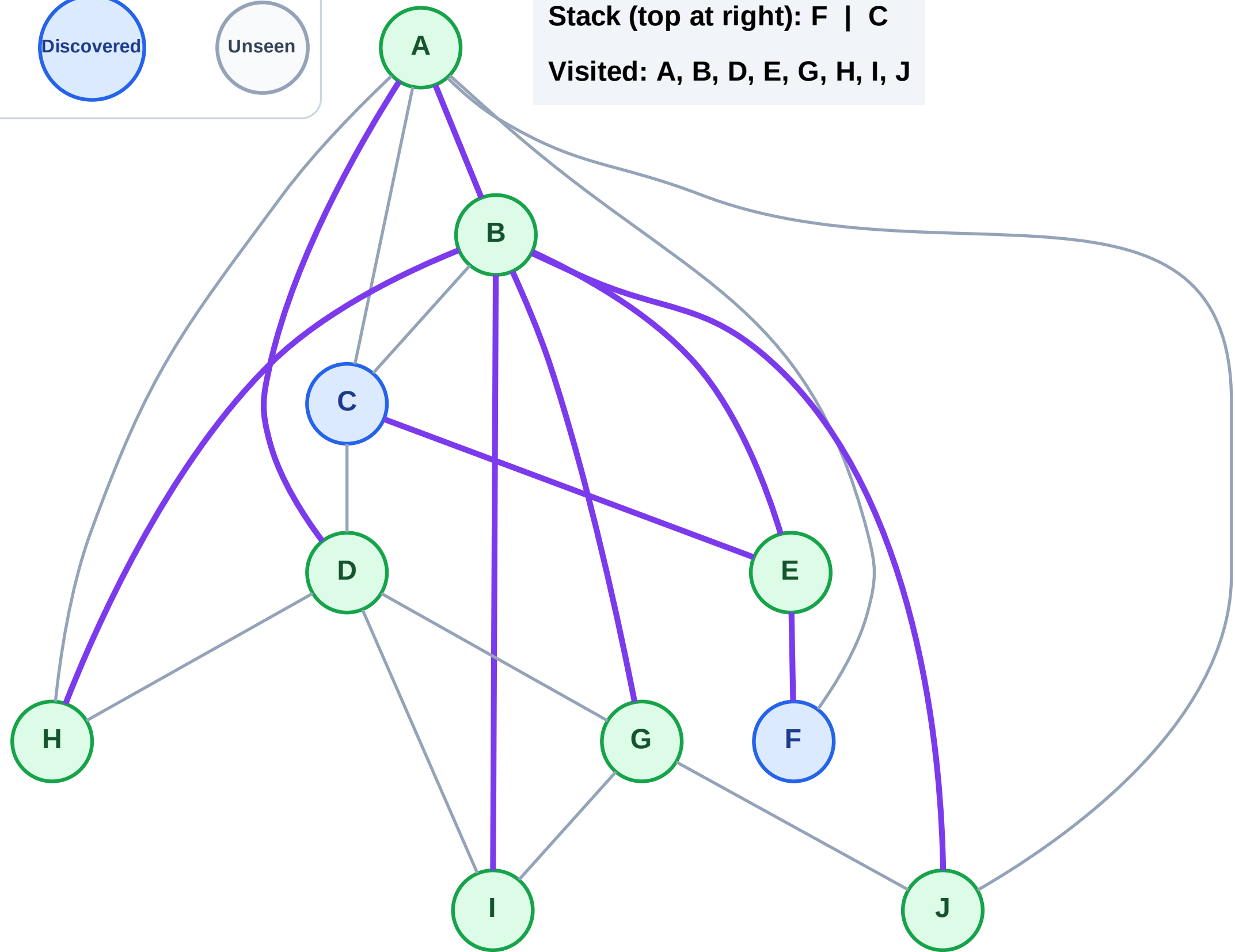
Step 17: No new neighbors from J

Legend



Stack (top at right): F | C

Visited: A, B, D, E, G, H, I, J



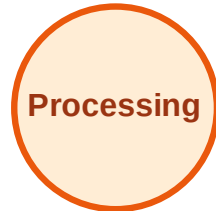
DFS Traversal

Step 18: Process node C

Legend



Complete



Processing



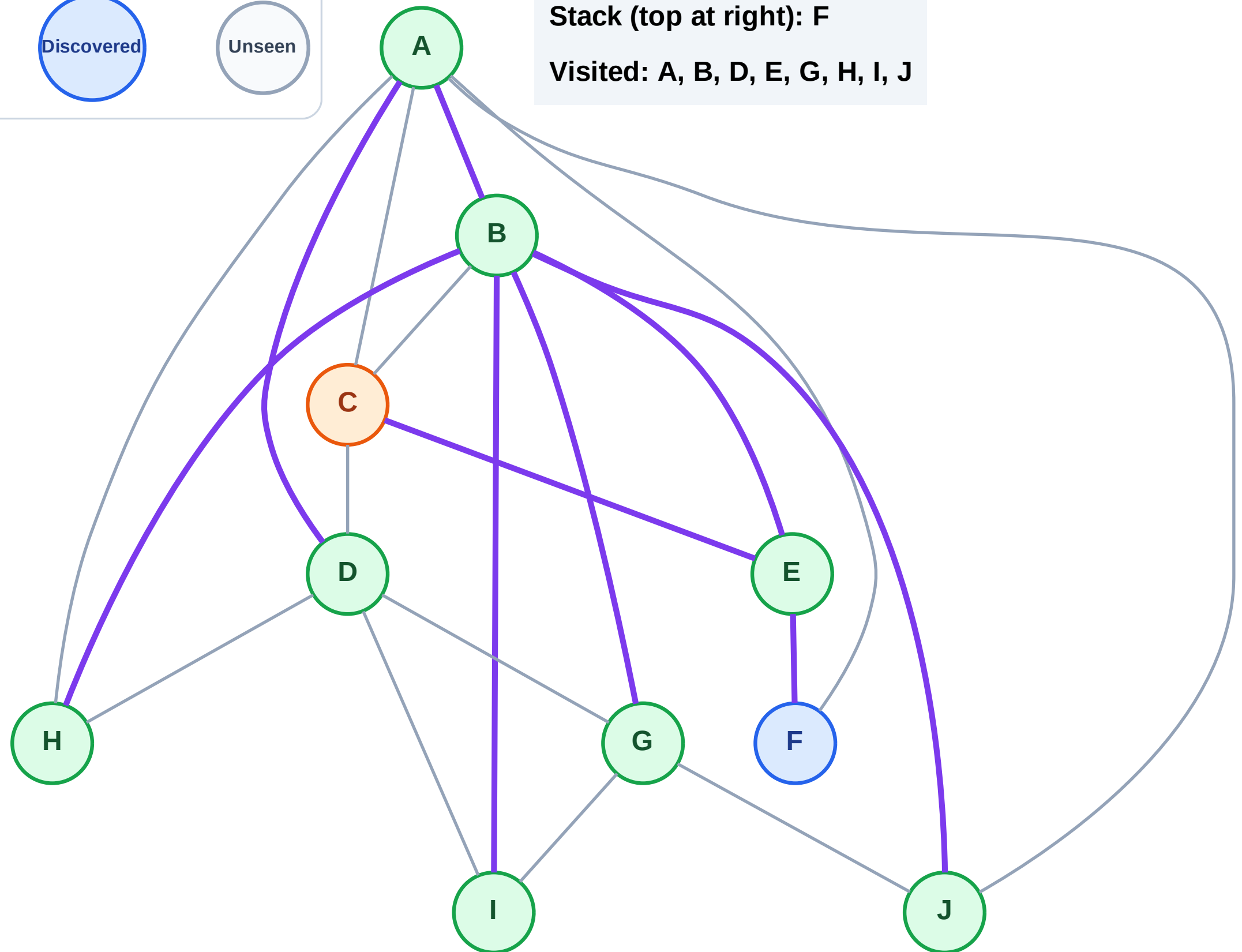
Discovered



Unseen

Stack (top at right): F

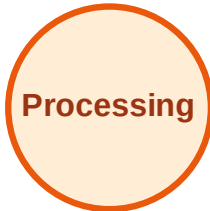
Visited: A, B, D, E, G, H, I, J



DFS Traversal

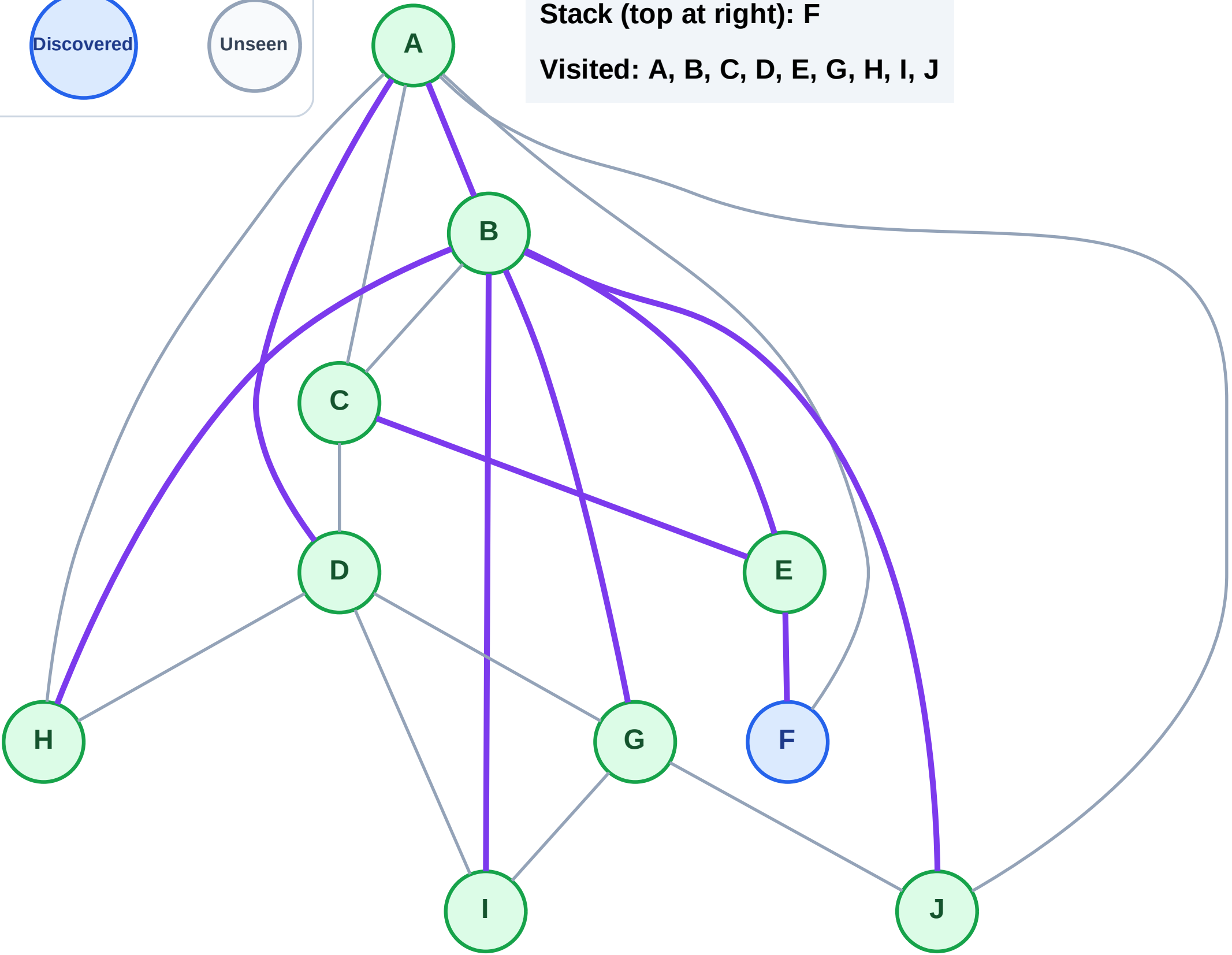
Step 19: No new neighbors from C

Legend



Stack (top at right): F

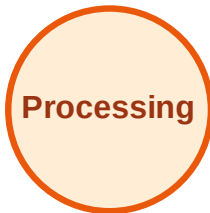
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

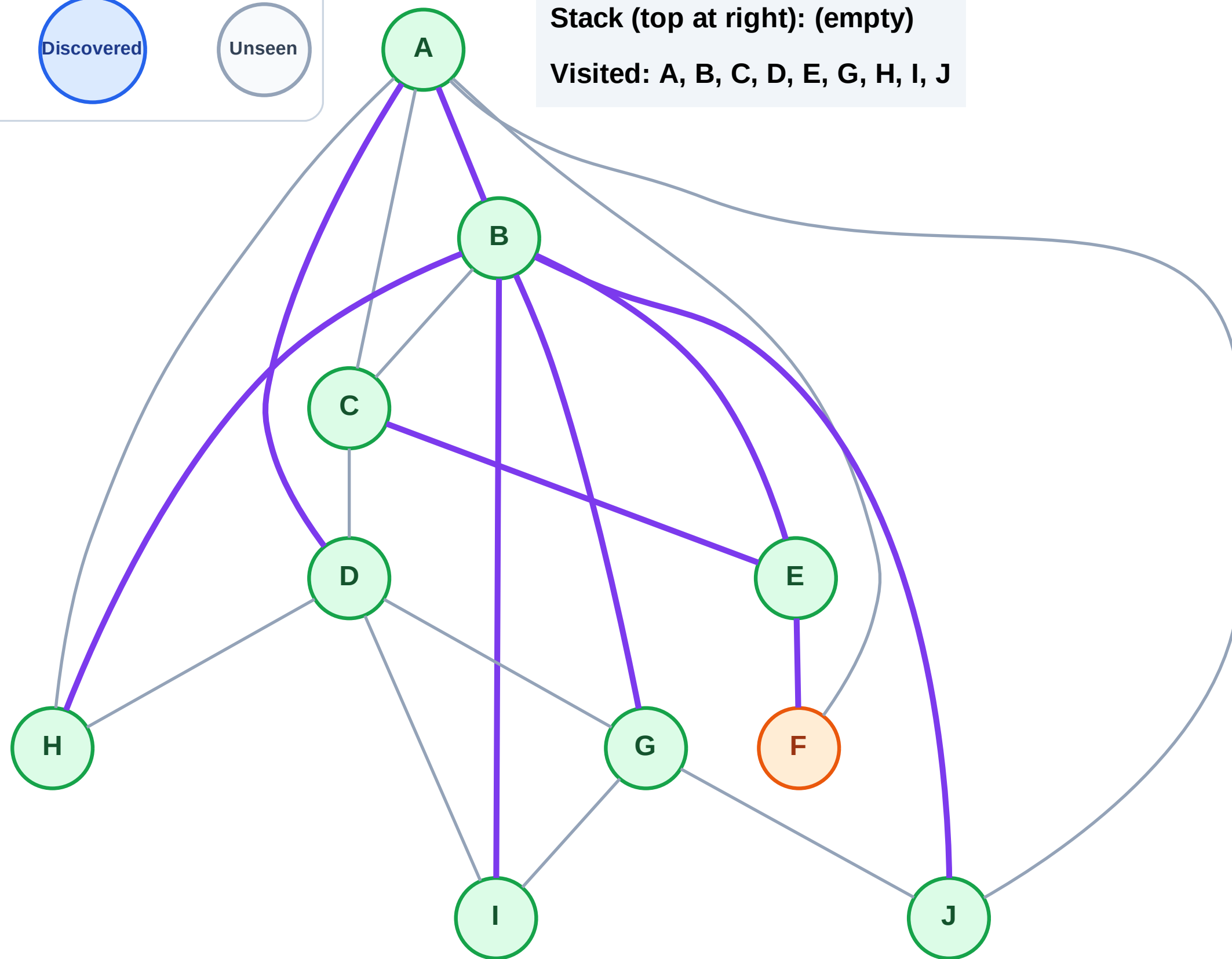
Step 20: Process node F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

